VALIDATION REPORT

HEGANG LONGYUAN WIND POWER CO., LTD.

VALIDATION OF THE

HEILONGJIANG WANGYUNFENG WIND POWER PROJECT

REPORT NO. CHINA-VAL/0231/2009

REVISION NO. 01

BUREAU VERITAS CERTIFICATION

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VALIDATION REPORT

Date of first issue: 20/12/2011	Organizational unit: Bureau Veritas Certification Holding SAS
Client: Hegang Longyuan Wind Power Co., Ltd.	Client ref.: Zhao Lijun

Summary:

Bureau Veritas Certification has made the validation of the Heilongjiang Wangyunfeng Wind Power Project owned by Hegang Longyuan Wind Power Co., Ltd. located in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, P.R.China on the basis of UNFCCC criteria for the CDM, as well as criteria given to provide for consistent project operations, monitoring and reporting. UNFCCC criteria refer to Article 12 of the Kyoto Protocol, the CDM rules and modalities and the subsequent decisions by the CDM Executive Board, as well as the host country criteria.

The validation scope is defined as an independent and objective review of the project design document, the project's baseline study, monitoring plan and other relevant documents, and consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final validation report and opinion. The overall validation, from Contract Review to Validation Report & Opinion, was conducted using Bureau Veritas Certification internal procedures.

The first output of the validation process is a list of Clarification and Corrective Actions Requests (CL and CAR), presented in Appendix A. Taking into account this output, the project proponent revised its project design document.

In summary, it is Bureau Veritas Certification's opinion that the project correctly applies the baseline and monitoring methodology ACM0002 version 12.1.0 and meets the relevant UNFCCC requirements for the CDM and the relevant host country criteria.

Report No.: CHINA-Val/0231/2009	Subject Group: CDM
Project title: Heilongjiang Wangyunfeng Wind Power Project	
Work carried out by: Li Yiting (Team Leader)	
Internal Technical Review carried out by: Wang Zhenning (Internal Reviewer)	

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Flavio Gomes

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Table of Contents *Page*

1 INTRODUCTION 4
1.1 Objective 4
1.2 Scope 4
1.3 Validation team 4

2 METHODOLOGY 5
2.1 Review of Documents 5
2.2 Follow-up Interviews 5
2.3 Resolution of Clarification and Corrective Action Requests 6
2.4 Internal Quality Control 6

3 VALIDATION CONCLUSIONS 7
3.1 Approval (49-50) 7
3.2 Participation (54) 8
3.3 Project design document (57) 8
3.4 Changes in the Project Activity 8
3.5 Project description (64) 9
3.6 Baseline and monitoring methodology 10
3.6.1 Baseline and monitoring methodology 10
3.6.2 Project boundary (80) 10
3.6.3 Baseline identification (87-88) 11
3.6.4 Algorithms and/or formulae used to determine emission reductions (92-93) 11
3.7 Additionality of a project activity (97) 13
3.7.1 Prior consideration of the clean development mechanism (104) 13
3.7.1.1 Historical information on project timeline 14
3.7.2 Identification of alternatives (107) 14
3.7.3 Investment analysis (114) 15
3.7.4 Barrier analysis (118) 20
3.7.5 Common practice analysis (121) 20
3.8 Monitoring plan (124) 21
3.9 Sustainable development (127) 22
3.10 Local stakeholder consultation (130) 22
3.11 Environmental impacts (133) 22

4 COMMENTS BY PARTIES, STAKEHOLDERS AND NGOS 23

5 VALIDATION OPINION 23

6 REFERENCES 24

7. CURRICULA VITAE OF THE DOE'S VALIDATION TEAM MEMBERS 28

APPENDIX A: CDM PROJECT VALIDATION PROTOCOL 29

1 INTRODUCTION

Hegang Longyuan Wind Power Co., Ltd. (hereafter called "**the PP**") has commissioned Bureau Veritas Certification to validate its CDM project Heilongjiang Wangyunfeng Wind Power Project (hereafter called "**the Project**") in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, P.R.China.

This report summarizes the findings of the validation of the project, performed on the basis of UNFCCC criteria, as well as criteria given to provide for consistent project operations, monitoring and reporting.

1.1 Objective

The validation serves as project design verification and is a requirement of all projects. The validation is an independent third party assessment of the project design. In particular, the project's baseline, the monitoring plan (MP), and the project's compliance with relevant UNFCCC and host country criteria are validated in order to confirm that the project design, as documented, is sound and reasonable, and meet the stated requirements and identified criteria. Validation is a requirement for all CDM projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of certified emission reductions (CERs).

UNFCCC criteria refer to Article 12 of the Kyoto Protocol, the CDM rules and modalities and the subsequent decisions by the CDM Executive Board, as well as the host country criteria.

1.2 Scope

The validation scope is defined as an independent and objective review of the project design document, the project's baseline study and monitoring plan and other relevant documents. The information in these documents is reviewed against Kyoto Protocol requirements, UNFCCC rules and associated interpretations.

The validation is not meant to provide any consulting towards the Client. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project design.

1.3 Validation team and Internal Technical Reviewer

The validation team and Internal Technical Reviewer consist of the following personnel:

FUNCTION	NAME	CODE HOLDER	TASK PERFORMED*
Team Leader	Ms. Li Yiting	☒**Yes** ☐ **No**	☒**DR** ☒**SV** ☒**RI**
Team Member	N. A.	☐**Yes** ☐ **No**	☐**DR** ☐**SV** ☐**RI**
Technical Specialist	N.A.	☐**Yes** ☐ **No**	☐**DR** ☐**SV** ☐**RI**
Financial Specialist	N.A.	☐**Yes** ☐ **No**	☐**DR** ☐**SV** ☐**RI**
Internal Technical Reviewer (ITR)	Mr. Wang Zhenning	☒**Yes** ☐ **No**	☒**DR** ☐**SV** ☐**RI**
Specialist supporting ITR	N.A.	☐**Yes** ☐ **No**	☐**DR** ☐**SV** ☐**RI**

*DR = Document Review; SV = Site Visit; RI = Report issuance

2 METHODOLOGY

The overall validation, from Contract Review to Validation Report & Opinion, was conducted using Bureau Veritas Certification internal procedures.

In order to ensure transparency, a validation protocol was customized for the project, according to the version 01.2 of the Clean Development Mechanism Validation and Verification Manual /Ref-1/, issued by the Executive Board at its 55th meeting on 30/07/2010. The protocol shows, in a transparent manner, criteria (requirements), means of validation and the results from validating the identified criteria. The validation protocol serves the following purposes:

- It organizes, details and clarifies the requirements a CDM project is expected to meet;
- It ensures a transparent validation process where the validator will document how a particular requirement has been validated and the result of the validation.

The completed validation protocol is enclosed in Appendix A to this report.

2.1 Review of Documents

The Project Design Document (PDD) submitted by Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd and additional background documents related to the project design and baseline, i.e. country Law, Guidelines for Completing the Project Design Document (CDM-PDD), Approved methodology, Kyoto Protocol, Clarifications on Validation Requirements to be Checked by a Designated Operational Entity were reviewed.

To address Bureau Veritas Certification corrective action and clarification requests Longyuan (Beijing) Carbon Asset Management Technology Co. Ltd. (the consultant) revised the PDD and resubmitted it on 01/08/2011 and the validation conclusion presented in this report relate to the project as described in the PDD version 02.

2.2 Follow-up Interviews

On 21/10/2009 Bureau Veritas Certification performed interviews with project stakeholders to confirm selected information and to resolve issues identified in the document review. Representatives of the PP, the consultant and local stakeholders were interviewed (see Section **6-References**). The main topics of the interviews are summarized in Table 1.

Table 1 Interview topics

Interviewed organization	Interview topics
Hegang Longyuan Wind Power Co., Ltd. (The PP)	✧ Project background information and CDM consideration. ✧ Project technology, operation, maintenance and monitoring capability. ✧ Project monitoring and management plan. ✧ Stakeholder consultation process. ✧ Project approval status (incl. EIA approval, CDM project approval status) ✧ Wind power development in the area ✧ Government policies related to wind power projects
Local Stakeholder	✧ Project background in details ✧ Stakeholder comments ✧ Social and environmental impact of the project
Longyuan (Beijing) Carbon Asset Management Technology Co. Ltd.	✧ Applicability of selected methodology. ✧ Baseline determination. ✧ Emission reductions calculation. ✧ Emission reduction monitoring plan.

2.3 Resolution of Clarification and Corrective Action Requests

The objective of this phase of the validation is to raise the requests for corrective actions and clarification and any other outstanding issues that needed to be clarified for Bureau Veritas Certification positive conclusion on the project design.

Corrective Action Requests (CAR) is issued, where:

(a) The project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions;

(b) The CDM requirements have not been met;

(c) There is a risk that emission reductions cannot be monitored or calculated.

Bureau Veritas Certification may also use the term Clarification Request (CL), if information is insufficient or not clear enough to determine whether the applicable CDM requirements have been met.

To guarantee the transparency of the validation process, the concerns raised are documented in more detail in the validation protocol in Appendix **A**.

2.4 Internal Quality Control

The validation report underwent an Internal Technical Review (ITR) before requesting registration of the project activity.

The ITR is an independent process performed to examine thoroughly that the process of validation has been carried out in conformance with the requirements of the validation scheme as well as internal Bureau Veritas Certification procedures.

The Team Leader provides a copy of the validation report to the reviewer, including any necessary validation documentation. The reviewer reviews the submitted documentation for conformance with the validation scheme. This will be a comprehensive review of all documentation generated during the validation process.

When performing an Internal Technical Review, the reviewer ensures that:

The validation activity has been performed by the team by exercising utmost diligence and complete adherence to the CDM rules and requirements.

The review encompasses all aspects related to the project which includes project design, baseline, additionality, monitoring plans and emission reduction calculations, internal quality assurance systems of the project participant as well as the project activity, review of the stakeholder comments and responses, closure of CARs, CLs and FARs during the validation exercise, review of sample documents.

The reviewer compiles clarification questions for the Team Leader and Validation Team and discusses these matters with Team Leader.

After the agreement of the responses on the 'Clarification Request' from the Team Leader as well as the PP(s) the finalized validation report is accepted for further processing such as uploading on the UNFCCC webpage.

3 VALIDATION CONCLUSIONS

In the following sections, the conclusions of the validation are stated.

The findings from the desk review of the original project design documents and the findings from interviews during the follow up visit are described in the Validation Protocol in Appendix A.

The Clarification and Corrective Action Requests are stated, where applicable, in the following sections and are further documented in the Validation Protocol in Appendix A. The validation of the Project resulted in 6 Corrective Action Requests and 9 Clarification Requests.

The CARs and CLs were closed based on adequate responses from the Project Participant(s) which meet the applicable requirements. They have been reassessed before their formal acceptance and closure.

The number between brackets at the end of each section corresponds to the VVM paragraph

3.1 Approval (49-50)

The letters of approval have been received and the following support documentation has been verified by Bureau Veritas Certification:

✍The DNA of China has issued a Letter of Approval (No. 2342) in Jan. 2010, authorizing Hegang Longyuan Wind Power Co., Ltd. as the Project Participant and confirms that the Project contributes to China's sustainable development /3/.

✍The DNA of UK has issued a Letter of Approval (EA/TotalG&PLtd/03/2011) on 04/10/2011, authorizing Total Gas & Power Ltd as the Project Participant for the Project /4/.

Bureau Veritas Certification received these letters of approval from the project participants and does not doubt the letters' authenticity.

The letters of approval do not contain a specific version of both the PDD and the validation report.

The title and contents of the letters of approval refer to the precise proposed CDM project activity title in the PDD being submitted for registration.

✌ Bureau Veritas Certification considers the letter of approval is in accordance with para. **45 - 48 /VVM**.

3.2 Participation (54)

The participation for the project participant has been approved by a Party of the Kyoto Protocol.

✌ Complying with para.**54/VVM**, Bureau Veritas Certification hereby confirms that by referring to the information on UNFCCC website i.e.

http://maindb.unfccc.int/public/country.pl?country=CN;

http://maindb.unfccc.int/public/country.pl?country=GB

3.3 Project design document (57)

✌ Complying with para.**57/VVM**, Bureau Veritas Certification hereby confirms that the PDD complies with the latest Project Design Document Form (CDM-PDD) version 03.2 and guidance documents for completion of PDD version 07.

3.4 Changes in the Project Activity

During the site visit, no physical changes were observed in the Project as compared to details mentioned in web-hosted PDD.

The major differences between these two versions of PDD are listed as bellow.

Items	**PDD version 01 (Webhosted)**	**PDD version 02**	**Validation Opinion**
1, Versions of Methodology and Tool to calculate the emission factor for an electricity system	ACM002, version 10 Tool to calculate the emission factor for an electricity system, version 1.1	ACM002, version 12.1.0 Tool to calculate the emission factor for an electricity system, version 02.2.1	ACM0002 version 12.1.0 was valid from 26/11/2010 and The “Tool to calculate the emission factor for an electricity system” version 02.2.1 was valid from 29/09/2011. Relevant part of the PDD has been updated. Please refer to CAR-4: and ~~CL-2:~~.
2, Input values of actual interest payable in the investment analysis and	The actual interest payable was not considered according to FSR. Project IRR without CER revenue is 6.26%	The actual loan from the bank was 25% of static total investments with the favourable rate of long-term loan about 5.346% were used in the IRR calculation	As a post tax benchmark is applied, the actual interest rate and equity-dept rate has been applied. It is in line with Guidelines on Assessment of Investment of Analysis. Please refer to CAR-6:.

Project IRR without CER revenue		based on the loan contract. Project IRR without CER revenue is 6.81%	
3, Common practice analysis	The standard to identify the similar projects are not clear.	The similar projects are identified as wind projects located in Heilongjiang Province after 2002. And statistics of wind power project in China in 2007, 2008 and 2009 were applied.	The revised common practice analysis is considered complete in accordance with the chosen standard and latest data source. Please refer to CL-9:.
4, Summary of the comments received from local stakeholders	The summary is not complete.	The revised summary of the comments received is consistent with the questionnaires.	The revised summary of the comments received is complete. Please refer to CL-6:.

3.5 Project description (64)

The Project is sited in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, P. R. China, which has central geographical coordinates of east longitude 130° 39′56″ (E130.66556°) and north latitude 47° 54′34″ (N47.90944°)

The total installed capacity of the Project is 49.3MW, consisting of 58 sets of 850kW wind turbines. The net electricity generated by the Project is 120,069MWh, will be sold to the Northeast China Power Grid (NECPG). As the NECPG is dominated by thermal power generation, the establishment of the Project is expected to achieve annual emission reductions of 123,430tCO_2e during the first seven years of its renewable crediting period.

The plant load factor (PLF) of 27.80% based on the approved FSR /5/ was conducted by a contracted authorized third party. Therefore, Bureau Veritas Certification confirms that the PLF defined in approved FSR complies with the requirement of “Guidelines for the Reporting and Validation of Plant Load Factors ver.1” (EB48, annex 11).

In the absence of the Project, equivalent amount of annual power output of the Project will be generated and supplied by Northeast China Power Grid (NECPG), which is dominated by the thermal power generation; this is same with the baseline scenario. The project scenario is considered additional in comparison to the baseline scenario, and therefore eligible to receive Certified Emissions Reductions (CERs) under the CDM, based on the analysis presented in the PDD.

The processes undertaken by Bureau Veritas Certification to validate the accuracy and completeness of the Project description include the document review and crosscheck with the reports and relevant approvals issued by local governments.

✌ The validation did not reveal any information that indicates that the Project can be seen as a diversion of official development assistance (ODA) funding towards the host country.

✌ Complying with para.**64/VVM**, Bureau Veritas Certification hereby confirms that the project description in PDD /2/ is accurate and complete in all respects and that there are no changes to the boundary as compared to the webhosted PDD.

3.6 Baseline and monitoring methodology

3.6.1 Baseline and monitoring methodology

The Project uses the approved consolidated baseline and monitoring methodology ACM0002 version 12.1.0– *"Consolidated baseline methodology for grid-connected electricity generation from renewable sources"* dated 17/09/2010./Ref-2/. The steps taken to assess the relevant information contained in the PDD against each applicability condition are described below.

By on-site visiting and interviewing with the PP, Bureau Veritas Certification confirms that the Project complies with the applicability conditions of methodology ACM0002 version 12.1.0:

- ✌ The Project is a grid-connected renewable wind power project that install a new power plant at a site where no renewable power plant was operated prior to the implementation of the Project (green-field plant);
- ✌ The Project does not involve switching from fossil fuels to renewable energy at the site of the Project.

Bureau Veritas Certification hereby confirms that the selected baseline and monitoring methodology, tool and other methodology component is previously approved by the CDM Executive Board, and is applicable to the Project, which, complies with all the applicability conditions therein.

Based on the on-site assessment, Bureau Veritas Certification hereby confirms that, as a result of the implementation of the Project, there are no greenhouse gas emissions occurring within the project boundary, which are expected to contribute more than 1% of the overall expected average annual emissions reductions, which are not addressed by the applied methodology.

3.6.2 Project boundary (80)

Bureau Veritas Certification has validated the project boundary by:

a) Assessing the relevant documents including FSR.

b) Observing the physical site and equipment used in the process.

The spatial extent of the project boundary is clearly defined in line with ACM0002 version 12.1.0 as the project power plant and all power plants connected physically to the NECPG that the Project is connected to. The greenhouse gases and emission sources included in the project boundary are CO_2 emissions from the electricity generation in fossil fuel fired power plants that are displaced due to the project activity.

✌ Complying with para.**80/VVM**, Bureau Veritas Certification hereby confirms that the identification of project boundary is in line with the delineation of grid boundaries as provided in the latest version of *"Notification on Determining Baseline Emission Factor of*

China's Grid" published by NDRC (China's DNA) on 02/07/2009 (hereafter called "*Notification of China-Grid EF*"). /10/ During on-site visit, via observations of the physical site, Bureau Veritas Certification hereby confirms that the identified boundary and the selected sources and gases are justified for the Project.

3.6.3 Baseline identification (87-88)

The steps taken to assess the requirement given in paragraph 81 and 82 of the VVM are described below:

The Project is the installation of a newly built and grid-connected renewable power plant that delivers the generated electricity to the NECPG, hence, according to methodology ACM0002, the baseline scenario is determined properly as:

Electricity delivered to the grid by the Project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the *"Tool to calculate the emission factor for an electricity system"* version 02.2.1 (hereafter called "*Tool-Grid EF*"). /Ref-3/

According to the "Notification of China-Grid EF", the delineation of grid boundary of the Project is the NECPG. Furthermore, the baseline of the Project determined in the PDD i.e. "electricity delivered to the grid by the Project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources within NECPG, as reflected in the combined margin (CM) calculations described" is transparent and deemed to be reasonable.

✌ Complying with para. **87 and 88/VVM**, Bureau Veritas Certification hereby confirms that:

(a) All the assumptions and data used by the project participants are listed in the PDD, including their references and sources;

(b) All documentation used is relevant for establishing the baseline scenario and correctly quoted and interpreted in the PDD;

(c) Assumptions and data used in the identification of the baseline scenario are justified appropriately, supported by evidence and can be deemed reasonable;

(d) Relevant national and/or sector policies and circumstances are considered and listed in the PDD;

(e) The approved baseline methodology has been correctly applied to identify the most reasonable baseline scenario and the identified baseline scenario reasonably represents what would occur in the absence of the proposed CDM project activity.

3.6.4 Algorithms and/or formulae used to determine emission reductions (92-93)

The steps taken to assess the requirement outlined in paragraph 89 the VVM are described below:

As per baseline methodology ACM0002 and "*Tool-Grid EF*", the baseline emission sources considered are the emission reduction ER_y during the crediting period is the difference between baseline emissions, project emissions and leakage. These are:

1) Baseline emissions: baseline emissions BE_y (tCO_2) are equal to baseline emission factor $EF_{grid,CM,y}$ (tCO_2/MWh) times the net electricity supplied to the grid EG_y (MWh).

2) Project Emissions: the project emissions are regarded as zero as per methodology ACM0002 version 12.1.0.

3) Leakage: no leakage has to be considered as per methodology.

4) Emission reductions:

$$ER_y = BE_y - PE_y = BE_y = EF_{grid,\ CM,\ y} \times EG_y = EF_{grid,\ CM,\ y} \times EG_{facility,y}$$

According to the baseline methodology ACM0002 Version 12.1.0 and "*Tool-Grid EF*" version 02.2.1, /Ref-3/ the baseline emission factor was calculated as six steps. In addition, the calculation in the PDD refers to the latest "*Notification of China-Grid EF*" published by China's DNA on 02/07/2009 /10/ which is most recent information available at the time of CDM-PDD submission to Bureau Veritas Certification for validation.

Bureau Veritas has checked the "Notification of China-Grid EF" and can confirm that the emission factor calculation is in accordance with data in the China Electric Power Yearbook from 2006 to 2008 and China Energy Statistical Yearbook from 2006 to 2008, and also complies with requirement the Tool-Grid EF. According to the Notification of China-Grid EF", the Simple OM emission factor ($EF_{grid,OM,y}$) of NECPG is calculated as 1.1293tCO_2e/MWh. Similarly, the build margin emission factor ($EF_{grid,BM,y}$) of the NCPG is calculated as 0.7242tCO_2e/MWh.

According to the "Tool-Grid EF", the default weights ω_{OM} = 0.75 for Operating Margin and ω_{BM} = 0.25 for build Margin in the first crediting period of Wind Power Projects are adopted.

Therefore, the combined baseline emission factor is determined ex-ante and will remain fixed during the first crediting period, viz.

$$EF_{grid,CM,y} = 1.1293 \times 0.75 + 0.7242 \times 0.25 = 1.0280 tCO_2e/MWh$$

According to the estimated annual electricity delivered to the grid 120,069MWh, the estimated annual emission reductions of the Project is 123,430tCO_2e during the first crediting period represents a reasonable estimation using the assumptions given by the Project.

✌ Complying with para.**92 and 93/VVM**, based on the above assessment, Bureau Veritas Certification hereby confirms that:

(a) All assumptions and data used by the project participants are listed in the PDD, including their references and sources;

(b) All documentation used by project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the PDD;

(c) All values used in the PDD are considered reasonable in the context of the proposed CDM project activity;

(d) The baseline methodology ACM0002 and "*Tool-Grid EF"* has been applied correctly to calculate project emissions, baseline emissions, leakages and emission reductions;

(e) All estimates of the baseline emissions can be replicated using the data and parameter values provided in the PDD.

3.7 Additionality of a project activity (97)

The steps taken and sources of information used, to cross-check the information contained in the PDD on this matter are described below:

"*Tool for Demonstration and Assessment of Additionality*" *version 5.2*.1 /Ref-4/ has been employed for demonstrating and assessing the additionality of the Project. The additionality of the Project has been carefully checked, in doing so Bureau Veritas Certification has put the main focus on the following issues:

3.7.1 Prior consideration of the clean development mechanism (104)

It has been demonstrated by the timeline of events of the Project that the CDM revenues were seriously considered in the decision to proceed with the Project prior to start of the Project.

From Table 2 bellow, Bureau Veritas Certification was able to verify that the start date of the Project determined as 12/05/2009 is appropriate (the signed date of the turbines purchase contract), which is the earliest of the dates at which the implementation or construction or real action of the Project began. This is in accordance with the latest CDM glossary.

The Project is a new project according to the definition in the "Guidelines on the demonstration and assessment of prior consideration of the CDM" version 05 (Annex 05, EB 62) (hereinafter called "Guidance-Prior Consideration") /Ref-5/, i.e. the start date of the Project is after 02/08/2008. The start date of the Project is before the date of publication of the PDD for global stakeholder consultation on 12/09/2009. Therefore, Bureau Veritas Certification verified the PP's notification to China's DNA and the UNFCCC secretariat in writing of their intention to seek CDM status on 02/07/2009 and 22/07/2009 respectively. They were both made within six months of the start date of the Project. Therefore it complies with *Guidance-Prior Consideration* /Ref-5/.

Bureau Veritas Certification has checked all reliable evidence from the PP, and is able to verify that all documents are substantial and authentic at that situation in the host country.

✌ According to the latest Glossary of CDM terms Ver. 05 /Ref-6/ and "*Guidance-Prior Consideration*", Bureau Veritas Certification confirms that the start date of the Project in the PDD is appropriate and reasonable at that situation.

✌ Complying with para.**100-103/VVM**, Bureau Veritas Certification has verified this issue, which could significantly influence the additionality of the Project, and confirms that the serious consideration under the context of the Project has been addressed appropriately in accordance with the above guidance. Consequently, the chronological events described with the relevant documented evidences are the objective foundation on which Bureau Veritas Certification developed its validation opinions.

3.7.1.1 Historical information on project timeline

Table 2 Timeline of the Project

Date	Events	Evidence verified
19/05/2008	EIA was finalized	/7/
25/07/2008	EIA was approved	/8/
Sep. 2008	FSR was finalized.	/5/
24/03/2009	FSR was approved	/6/
02/04/2009	Project owner made the investment decision based on the CDM consideration	/11/
Apr. 2009	Consultancy contract to start CDM development was signed	/12/
May 2009	Stakeholders questionnaires were finished	/51/
12/05/2009	The wind turbines purchase agreement of was signed between the PP and manufacturer	/15/
14/05/2009	The transformer purchase agreement of was signed between the PP and manufacturer	/16/
04/06/2009	Construction Start Permission was issued	/18/
05/06/2009	The construction contract was signed	/17/
02/07/2009	The PP's notification to China's DNA in writing of their intention to seek CDM status was handed in and the confirm letter was received on 19/08/2009	/14/
22/07/2009	The PP informed UNFCCC secretariat in writing of the commencement of the Project and their intention to seek CDM on 20/07/2009 and received the confirm letter on 22/07/2009	/13/

3.7.2 Identification of alternatives (107)

The plausible and credible alternatives to the Project were identified as per the "*Tool-Additionality*" Version 05.2.1 /Ref-4/:

Alternative (a): The Project not undertaken as a CDM project activity;

Alternative (b): Thermal power plant with the same annual capacity output as the Project;

Alternative (c): Other renewable energy project with the same annual electricity output as the Project;

Alternative (d): To provide the same electricity output by the NECPG.

Bureau Veritas Certification confirms the listed alternatives to be credible and complete.

Alternative (b) was correctly eliminated through examination of the laws or regulations in China. According to the Notice on Strictly Prohibiting the Installation of Thermal Generators with the Capacity of 135MW or below issued by the General Office of the

State Council, Decree No. [2002]6 /20/, construction of thermal power plants less than 135MW are prohibited in the areas covered by the large grid such as provincial grids in China.

Alternative (c) was eliminated by analyzing the feasibility of development of local renewable energy resources including solar PV, geothermal, biomass and hydropower. Realizing the technology development status and the high cost of power generation from solar PV, geothermal and biomass, solar PV and geothermal with equivalent output is unfeasible in China /21//22//23/. Furthermore, due to lack of resources of hydropower in project area /24/, alternative (c) is excluded.

✌ Complying with para.**107/VVM**, Bureau Veritas Certification was able to verify that the alternatives identified to the Project are credible and complete, and found satisfactory to exclude alternatives (b) and (c). Hence **Step 1** of "*Tool-Additionality*" was applied appropriately.

3.7.3 Investment analysis (114)

Considering the baseline scenario identified above, option III, the Benchmark Analysis, is applied in the investment analysis as per the *Sub-step 2b* of **Step 2** of "*Tool-Additionality*", which is in accordance with "*Guidelines on the Assessment of Investment Analysis*" (Ver. 05) /Ref-8/.

Project IRR of 8% (post-tax) was employed by the Project as benchmark, which is sourced from the "Interim Rules on Economic Assessment of Electric Power Engineering Retrofit Projects" /25/ issued by State Power Corporation of China in 2002. Bureau Veritas Certification has verified this benchmark and confirms that it is widely applied in Chinese power generation industries; therefore, Bureau Veritas Certification confirms that the benchmark is suitable for the Project.

Before reviewing the IRR calculation, Bureau Veritas Certification has validated the basic parameters listed in the PDD in accordance with the **Para. 113/VVM** /ref-1/.

According to the relevant evidence provided, Bureau Veritas Certification confirms that the PP's final decision to proceed with the investment in the Project has been made based on the FSR /6/, which was finalized in **Sep. 2008** and approved on **24/03/2009**. Later, based on the conclusion in the FSR, PP decided to invest the Project on **02/04/2009** /11/ with CDM consideration. Given this relative short period of time between the FSR and the decision to proceed with the Project, Bureau Veritas Certification was therefore confident that it is unlikely in the context of the underlying Project that the input values would have materially changed, which is in line with the **Para. (a) 113/VVM**.

At the same time, Bureau Veritas Certification compared the input values for the financial analysis in the PDD and the approved FSR, as the interest payable was not considered to calculate income tax in the FSR, Bureau Veritas Certification confirms that except the actual interest payable is considered, all the other parameters are consistent with the approved FSR. As a post tax benchmark is applied, the actual interest payable should be taken into account according to "*Guidelines on the assessment of investment analysis*" version 05. Bureau Veritas Certification was therefore with opinion of that the investment analysis is in accordance with **Para. (b) 113/VVM**.

Furthermore, Bureau Veritas Certification has reviewed the IRR calculation sheet and confirms that:

- The **operation period** of 20 years were selected reasonably following the requirements of *"Interim Rules on Economic Assessment of Electric Power Engineering Retrofit Projects"* /25/.
- The **static total investment** in the approved FSR is 482.69 million RMB (about 9,791RMB/kW, 979€/kW).
  - According to China Wind Power Report 2008 published by China Environmental Science Press in Sep. 2008 /26/, the unit cost of wind power projects varies from 800€/kW to 1150€/kW, and the investment of the Project falls in this range and was verified appropriate.
  - Bureau Veritas Certification has also checked available information for registered wind projects public in UNFCCC in Heilongjiang Province (table 3), and found that the unit static investment vary from 8,276 RMB/kW to 11,186 RMB/kW. The unit static investment of the Project is calculated to be 9,791RMB/kW, within the range above. Therefore, Bureau Veritas Certification is able to confirm that the total static investment used is appropriate and reasonable.
  - Bureau Veritas Certification has also checked the Final Accounting Report of the Project /54/, the actual static total investment of the Project is 489 million, which is 1.3% higher than the estimated one in the FSR. Therefore, Bureau Veritas Certification is able to confirm that the total static investment used is appropriate and reasonable.

Table 3 Input values of the registered wind projects public in UNFCCC in Heilongjiang Province

Ref No.	Project	Installed Capacity (MW)	static investment (RMB/kW)	Unit Annual O&M cost (RMB/kWh)	PLF
0829	Yichun Daqingshan Wind Power Project	16.15	10176	0.217	24.51%
0906	Heilongjiang Huafu Muling Wind Farm Project	31.2	**11186**	0.138	25.49%
0969	Yichun Erduoyan Wind Power Project 28.05 MW	28.05	9231	**0.225**	22.96%
1147	Yichun Shimaodingzi Wind Power Project 30.6MW	30.6	9834	0.221	24.48%
1209	Wuerguli 30 MW Wind Power Project	30	10924	**0.096**	**26.03%**
1310	Guohua Qiqihaer Fuyu 1st Stage Wind Farm Project	49.5	8471	0.105	23.13%
1816	Heilongjiang Shiwenzi WPP	49.5	11015	0.329	24.64%
2032	Heilongjiang Dajiazishan 49.3MW WPP	49.5	9668	0.127	22.23%
2035	Heilongjiang Yilan Maanshan WPP	49.5	8361	0.145	23.63%
2049	Heilongjiang Beiantun 49.3MW WPP	49.5	9410	0.128	**21.72%**

2056	Huanan hengdaishan East WPP	24.65	8809	0.183	23.76%
2200	Heilongjiang Huanan Hengdaishan West WPP	45.05	**8276**	0.157	23.64%
2777	Heilongjiang Shaobaishan Wind Power Project	49.5	9399	0.111	25.97%

- The **electricity tariff** applied in the PDD is 0.61RMB/kWh (incl. VAT) in the first 30,000 operation hours and 0.41RMB/kWh (incl. VAT) after 30,000 operation hours.

  The Project is located in Heilongjiang Province. Bureau Veritas Certification has studied the tariff notifications for wind power projects located in Heilongjiang Province and summarized as follows:

- After the Chinese power sector reform since April 2002 /27/, on 10/12/2002, NDRC issued the document Ji Ji Chu [2002] No. 2692 /28/ and two-phase tariff structure was approved in this document.

- On 04/01/2006, China's government issued the Tentative Management Measures for Price and Sharing of Expenses for Electricity Generation from Renewable Energy (Document No. Fa Gai Jia Ge [2006]7) /29/. After that, the tariff of most wind power projects began to be approved by government.

- On 23/07/2008, National Development and Reform Commission issued the Notice on consolidating the Feed-in Tariff of Wind Power Project (Code: Fa Gai Jia Ge [2008] 1876) /30/, and the tariff of wind power projects located in Heilongjiang Province for the first 30,000 operation hours was approved as 0.61RMB/kWh (incl. VAT), which equals to 0.61RMB/kWh (incl. VAT), in the document. After the first 30,000 operation hours, the tariff was approved as the local average on-grid tariff.

- On 29/06/2008, National Development and Reform Commission issued the Notice on adjustment of tariff in NECPG (Code: Fa Gai Jia Ge [2008] 1678) /31/, and the local average on-grid tariff was 0.3369RMB/kWh (excl. VAT). The applied tariff of 0.41RMB/kWh (incl. VAT) after 30,000 operation hours is 12% higher than the local average on-grid tariff thus is considered conservative.

The FSR was finalized in Sep. 2008, and board decision was made on 02/04/2009, at which the Fa Gai Jia Ge [2008] 1876 and Fa Gai Jia Ge [2008] 1678 were available after the implementation of Renewable Energy Law. Therefore, Bureau Veritas Certification can confirm that 0.61RMB/kWh (incl. VAT) in the first 30,000 operation hours and 0.41RMB/kWh (incl. VAT) after 30,000 operation hours used in investment analysis is available and conservative at the time of board decision. It is also evidenced by the Power Purchase Agreement of the Project /53/.

Furthermore, by checking the Information Note on the Highest Tariffs Applied by the EB in Its Decision on Registration of Projects in the People's Republic of China (Version 2) published by EB /32/, the highest tariff in Heilongjiang Province is 0.61RMB/kWh (incl. VAT) for the first 30,000 hrs and after that tariff for thermal power plants will be applied. The project is still additional with the highest tariff of Heilongjiang Province.

Therefore, Bureau Veritas Certification confirms that the tariff of 0.61RMB/kWh (incl. VAT) in the first 30,000 operation hours and 0.41RMB/kWh (incl. VAT) after 30,000 operation hours employed in investment analysis is appropriate.

- The **annual electricity generated** of the Project in the investment analysis was based on the 30-year long term wind resource history data of the local area and one year onsite measured data in 2006, and WAsP software as stated in approved FSR. Therefore the supplied electricity is found appropriate.
  - The annual operation hour of 2,435.47h was based on the approved FSR, which was conducted by a third party contracted with the PP. Therefore, Bureau Veritas Certification confirms that the plant load factor defined in the FSR complies with the requirement of "Guidelines for the Reporting and Validation of Plant Load Factors ver.1" (EB48, annex 11) /Ref-7/.
  - According to China Wind Power Report 2008 /26/ published by China Environmental Science Press in Sep. 2008, annual utilization hours of wind power projects varies from 1,325h to 2,401h, and the average annual utilization hours of wind power projects in China is 1,787h. The annual utilization hours of the Project are higher than the average value, hence was verified conservative.
  - Bureau Veritas Certification has also checked available information for registered wind projects public in UNFCCC in Heilongjiang Province (table 3), and found that the PLF vary from 21.72% to 26.03%. The PLF of the Project is 27.80%, which is higher than the range above. Therefore, Bureau Veritas Certification is able to confirm that the annual output used is conservative.

- Bureau Veritas Certification confirms that the **annual O&M cost** is the sum of salary and welfare of employees, materials fee, insurance cost, maintenance fee and miscellaneous account, which was studied based on the "Code on Compiling Feasibility Study Report of Wind Farms" issued by NDRC /34/ and "Economic Evaluation Method and Parameters for Project Construction" (versiorn 3) /35/.
  - Bureau Veritas Certification has also checked available O&M cost information for registered wind projects public in UNFCCC in Heilongjiang Province (table 3 above), and found that the unit O&M cost vary from 0.096 RMB/kW to 0.329 RMB/kW. The unit O&M cost of the Project is calculated to be 0.117 RMB/kWh, within the range above. Therefore, Bureau Veritas Certification is able to confirm that the annual O&M cost used is appropriate.

- A post-tax benchmark is applied for the investment analysis of the Project. Bureau Veritas Certification has checked the IRR calculation sheet and confirms that the interest has been taken into account in the calculation of income tax. Furthermore, Bureau Veritas Certification has checked the loan contract signed between the PP and the bank /38/ on 28/03/2009, and found that the Project actually enjoys 5.346% of the **interest rate** and the dept rate is 25%. As a post tax benchmark is applied, the actual interest payable should be taken into account according to "*Guidelines on the assessment of investment* *analysis*" version 05. Bureau Veritas Certification confirms that the interest payable in the calculation of income tax is appropriate. 5.346% is 90% discount of 5.94% which is the same as prevailing commercial interest rates in China /39/.

- The **residual value rate** of 5% was selected reasonably following relevant regulation in China, i.e. The Document of the State Administration of Taxation, Decree No. 70 of Guoshuifa [2003] on 30 June 2003 /40/ and Statute of Income Tax Law of China issued on 06/12/2007 /41/.
- Bureau Veritas Certification has checked the IRR calculation sheet and confirm that **depreciation** has been deducted in estimating gross profits on which tax is calculated, and be added back to net profits for the purpose of calculating the financial indicator. Bureau Veritas Certification confirms that the depreciation calculated complies with "Economic Evaluation Method and Parameters for Project Construction" (version 3), which regulated that that the annual depreciation value can be calculated as fixed value minus residue value and then multiply the depreciation rate, the calculation method in the IRR sheet is calculated correctly follow this equation. And the depreciation of 15 years is in compliance with 'the minimum years, i.e. 10 years, for computing depreciation of fixed assets for machinery or mechanical apparatuses as per Implementation regulation on the Enterprise Income Tax Law issued on 06/12/2007/41/.
- Bureau Veritas Certification has also verified values of various **taxes** through crosschecking with the taxation rules conducted by local government and found to be fully consistent.
  - The VAT of 8.5% means half of the normal VAT of 17% will be imposed for wind power projects. It complies with the *Notice of Value added Tax Policy Regarding Products Using Certain Synthesized Resources and Other Products [2001] No. 198* issued by the Ministry of Finance and the State Administration of Taxation in Dec. 2001/42/ at the time of FSR was finished.
  - The income tax of 25% complies with Enterprise Income Tax Law of China which is effective from 01/01/2008 /43/.

In summary, based on the above reliable data sources, Bureau Veritas Certification was able to confirm that the actual interest payable from the loan contract and the other input values from the approved FSR are valid and applicable at the time of FSR was finished. Therefore, Bureau Veritas Certification confirms that the investment analysis is in accordance with **Para. (c) 113/VVM**.

The project IRR of the Project without CDM revenues is 6.81%, much lower than the benchmark, which shows that the Project is not financially attractive compared to the benchmark in the absence of CDM benefits.

Bureau Veritas Certification has reviewed the IRR calculation /44/ and confirms that the IRR processing is consistent with the "*Guidelines on the assessment of investment analysis*" version 05 and the data sources as well as the analysis approach are reliable and based on the FSR linking directive to the actual situation of the host country. Four financial parameters were taken as uncertain factors for sensitive analysis of financial attractiveness. The sensitivity analysis is based on the initial project IRR in PDD which is the basis of the investment decision

- Static total investment
- Annual O&M cost

- Annual electricity generated
- Tariff (excluding VAT)

According to "Code on compiling feasibility study report of wind power projects" published by NDRC /34/, total static investment, annual grid connected output and tariff should be taken as uncertain factors to do sensitivity analysis, and ±10% variation of above factors shall be considered in the sensitivity analysis. Therefore, Bureau Veritas Certification confirms that the variables and variations ±10% performed for sensitivity analysis is deemed to be appropriate for the Project.

- With a decrease in **static total investment** by 7.80%, the Project IRR may reach 8%. However, it has been verified that the actual contracts of wind turbines /15/, and found that the total value of the contracts is 306.30 million RMB and is 9.00% higher than the estimated values in the FSR of 281.01 million RMB. And according to the Final Accounting Report of the Project /54/, the actual static total investment of the Project is 489 million, which is 1.3% higher than the estimated one in the FSR. Bureau Veritas Certification is confident that the total static investment won't decrease by 7.80%.
- The **annual O&M cost** comprises materials expense, maintenance cost, employee salary and welfare and miscellaneous accounts. With a decrease in annual O&M cost by 36.30%, the Project IRR may reach 8%. However, it is evidently impossible as the cost increasing in resent year /47/.
- With an increase by 8.18% in **annual electricity generated**, the project IRR will reach the benchmark. By checking the FSR, the supplied electricity of the Project is based on wind resource data of 30-year and WAsP software as stated in the FSR. Therefore, Bureau Veritas Certification confirms that it is unlikely that the supplied electricity could increase by 8.18% during the whole life of the Project.
- With an increase of **tariff** by 8.18%, the Project IRR will reach 8%. Bureau Veritas Certification has confirms the highest tariff of wind power projects located in Heilongjiang Province has been guided by NDRC at 0.61RMB/kWh (incl. VAT) which is also the highest tariff in Heilongjiang Province issued by EB/32/. Therefore, Bureau Veritas Certification concludes that it is unlikely that the tariff could increase to make IRR reach benchmark of 8.18%.

Considering of the CERs sales revenues (calculated with 10.5 EUR/tCO_2e), the project IRR of the Project can be crossing the benchmark at 10.39%.

Bureau Veritas Certification can conclude that both of the variation range and relevant assumptions stated in the PDD are robust and the investment of the Project is deemed to be financially unattractive.

✌ Complying with para.**114/VVM**, Bureau Veritas Certification hereby confirms that the underlying assumptions are appropriate and the financial calculations are correct.

3.7.4 Barrier analysis (118)

The **Step 3** Barrier analysis was not applied for the Project.

3.7.5 Common practice analysis (121)

The Common practice analysis was addressed as per **Step 4** of "*Tool-Additionality*" and latest rules issued by EB.

The Project is a newly built 49.3MW wind power project in the area of Heilongjiang Province. Therefore, the project activities similar to the Project should be the wind power projects located in Heilongjiang Province.

The Power System reform happened in 2002 in China /27/. New power project, including wind power, was expected to compete under more commercial conditions.

Following these criteria, Bureau Veritas Certification has verified the wind power projects as identified in the PDD by crosschecking the public statistics i.e. "*Statistics of wind power installed capacity in China*" *(2007, 2008, 2009)* /48/, and two wind power projects (Huanfu Mulan Wind Farm and Huanfu Fujin Wind Farm) are identified as per the above criteria.

As the two identified similar projects were demonstration projects and funded by international low interest loan /49/ or national soft loan /50/. Bureau Veritas Certification has verified the description in the PDD and found that it is consistent with the sectoral statistics and therefore can conclude that the Project is not common practice in the region.

✌ Complying with para.**121/VVM**, Bureau Veritas Certification has verified the description in the PDD and found that it is consistent with the sectoral statistics and therefore confirms that the Project is not common practice in the region.

Based on above demonstration that in accordance with "*Tool-Additionality*" and supported by reliable data sources, it is the opinion of Bureau Veritas Certification that the Project is thus additional.

3.8 Monitoring plan (124)

Bureau Veritas Certification hereby confirms that the monitoring plan complies with the requirements of the methodology.

The steps taken to assess whether the monitoring arrangements described in the monitoring plan are feasible within the project design are described below.

The Project uses the approved consolidated monitoring methodology ACM0002 version 12.1.0 for grid connected electricity generation from renewable sources.

Applicability of this methodology is justified in PDD as it involves grid connected renewable power generation using wind energy. Refer discussions on the validity of the methodology at Section 3.6.1 above. Bureau Veritas Certification hereby confirms that the monitoring plan complies with the requirements of the methodology.

The ex-ante combined margin emission factor is determined based on the most recent information available. According to the monitoring plan, the electricity exported to the grid and electricity utilized by the Project will be monitored by two bidirectional meters (accuracy is not less than 0.5%) installed on the project site. And the net electricity supplied to the grid is the difference of electricity exported to the grid and utilized by the Project. The electricity will be continuously measured and recorded monthly. Data may be verified against by sales receipts. The meters are expected to be calibrated annually. Bureau Veritas Certification is of the opinion that the monitoring plan complies with the requirements of the methodology.

Operational management for the project activity is comprehensively detailed in PDD and this includes description of the responsibility, procedure reference, calibration frequency and maintenance needs.

By on-site interviewing with the PP, Bureau Veritas Certification confirms that the monitoring arrangements described in the monitoring plan are feasible within the project design, and the means of implementation of the monitoring plan are sufficient to ensure the emission reductions achieved by the Project can be reported ex post and verified.

✌ Complying with para.**124/VVM**, Bureau Veritas Certification hereby confirms that the monitoring arrangements described in the monitoring plan are feasible within the project design and the project participants are able to implement the monitoring plan.

3.9 Sustainable development (127)

China's DNA confirmed the contribution of the Project to the sustainable development of the host Party. Refer to item 3.1 of this report.

3.10 Local stakeholder consultation (130)

Prior to the publication of the PDD on the UNFCCC website, viz, in May 2009, the PP conducted surveys to stakeholders including local residents and governments. 10 stakeholder representatives were invited in a symposium on 06/05/2009 and all the stakeholder representatives support and welcome the proposed project. Then total 40 questionnaires had been distributed in nearby areas.

According to the 40 filled questionnaires, the outcome of this survey shows that the interested stakeholder have a very good understanding of the Project, and they agreed that the Project can increase the employment opportunities, reduce air pollution, improve living standards and increase income. 100% of the local stakeholders supported the construction of the Project.

The stakeholders have recognized the contribution of the Project to local environment and social economy. Their views were endorsed by the local stakeholders interviewed during the site visit of the validation activity.

During the on-site visit, Bureau Veritas Certification has conducted an interview with local stakeholders and confirms that the stakeholders impacted had been invited in a transparent manner. The interview with stakeholders and review of returned questionnaires shows that the summary of the comments received has been completely provided in the PDD and due account of the comments has been described in the PDD. Bureau Veritas Certification hereby confirms that the process of local stakeholder consultation is observed to be adequate.

✌ Complying with para.**130/VVM**, Bureau Veritas Certification hereby confirms that the local stakeholder consultation was performed and the process of local stakeholder consultation is observed to be adequate. The Project will be beneficial to the local sustainable development without negative affect on the local stakeholders.

3.11 Environmental impacts (133)

The PP have undertaken an analysis of environmental impacts and Bureau Veritas Certification confirms that the Environmental Impact Assessment was carried out by the qualified entity, and approved by the Environmental Protection Adminitration of Heilongjiang Province on 25/07/2008 (Code: Heihuanjianshen [2008] No.164) /8/.

The environmental impact caused by the Project has been identified and analyzed in the PDD. By checking the EIA report, Bureau Veritas Certification is able to ensure that the environment impact occurs mainly in the construction period due to waste water, dust and exhaust gas, noise pollution, solid waste, and ecological deterioration. All above impacts would be within an acceptable limit by implementing corresponding mitigation measures as per the statement of the EIA. The impacts mentioned above were insignificant according to the conclusion of the EIA.

✌ Complying with para.**133/VVM**, Bureau Veritas Certification hereby confirms that the Project will not have any significant impacts on the environment by means of measures of pollution avoidance and control as well as ecological recovery.

4 COMMENTS BY PARTIES, STAKEHOLDERS AND NGOS

✌ Complying with para.**173/VVM**, the PDD using methodology ACM0002 was webhosted on the UNFCCC for global stakeholders' comments as per CDM requirements. The Project was webhosted from 12/09/2009 to 11/10/2009.

No comments were received during this period.

5 VALIDATION OPINION

Bureau Veritas Certification has performed a validation of the Heilongjiang Wangyunfeng Wind Power Project in P. R. China. The validation was performed on the basis of UNFCCC criteria and host country criteria and also on the criteria given to provide for consistent project operations, monitoring and reporting.

The validation consisted of the following three phases: i) a desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) the resolution of outstanding issues and the issuance of the final validation report and opinion.

Project participants used the latest *Tool for demonstration and assessment of additionality* (version 05.2.1), to demonstrate the additionality of the Project. In line with this tool, the PDD provides investment analysis to determine that the project activity itself is not the baseline scenario. The latest *Tool to calculate the emission factor for an electricity system* (version 02.2.1) is also applied to determine the emission factor of Northeast China Power Grid.

By synthetic description of the project, the Project is likely to result in reductions of GHG emissions partially. An investment analysis demonstrates that the project activity is not a plausible baseline scenario. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity. Given that the project is implemented and maintained as designed, the project is expected to achieve the average annual emission reductions of 123,430tCO_2e over the chosen 7-year renewable crediting period.

The review of the project design documentation (version 02) and the subsequent follow-up interviews have provided Bureau Veritas Certification with sufficient evidence to determine the fulfillment of stated criteria. In our opinion, the project correctly applies and meets the relevant UNFCCC requirements for the CDM and the relevant host country criteria. Bureau Veritas Certification thus requests registration of Heilongjiang Wangyunfeng Wind Power Project as CDM project activity.

6 REFERENCES

Category 1 Documents:

Documents provided by the consultant that relate directly to the GHG components of the project.

/1/	PDD version 01 dated 26/08/2008 http://cdm.unfccc.int/Projects/Validation/DB/4JTBKWDX499I7O59PJ3TUTVO5ZEUSD/view.html
/2/	PDD version 02 dated 01/08/2011
/3/	LoA from DNA of China (Host country) in Jan. 2010 (No. 2342)
/4/	LoA from DNA of UK on 04/10/2011 (EA/TotalG&PLtd/03/2011)
/5/	Feasible Study Report (FSR) completed by China Fulin Windpower Development Corporation in Sep. 2008
/6/	Feasibility Study Report (FSR) approval issued by Development and Reform Commission of Heilongjiang Province on 24/03/2009 (Code: Hei Fa Gai Wai Zi Neng Yuan [2009] No. 302)
/7/	Environment Impact Assessment Report (EIA) completed on 19/05/2008
/8/	EIA approval issued by Environmental Protection Adminitration of Heilongjiang Province on 25/07/2008 (Code: Heihuanjianshen [2008] No.164)
/9/	National Renewable Energy Law issued by NDRC of China effective from 01/01/2006. http://www.windpower.org.cn/news/links/fl_2005_0510_02.htm
/10/	Notification on Determining Baseline Emission Factor of China's Grid dated 02/07/2009.
/11/	Board investment decision made on 02/04/2009
/12/	CDM consultation service contract signed in Apr. 2009
/13/	Notice to inform the UNFCCC secretariat of intention to seek CDM status confirmed on 22/07/2009 http://cdm.unfccc.int/Projects/PriorCDM/notifications/index_html
/14/	Notice to inform the NDRC of intention to seek CDM status handed in 02/07/2009 and the confirm letter was received on 19/08/2009
/15/	The wind turbines purchase agreement wind turbines between the PP and manufacturer signed on 12/05/2009
/16/	The transformer purchase agreement wind turbines between the PP and manufacturer signed on 14/05/2009
/17/	The construction contract was signed on 05/06/2009
/18/	Construction Start Permission on 04/06/2009
/19/	Main transformer purchase contract signed on 01/06/2009

/20/	Notice on Strictly Prohibiting the Installation of Thermal Generators with the Capacity of 135MW or below issued by the General Office of the State Council, Decree No. 2002 6. http://www.gov.cn/gongbao/content/2002/content_61480.htm
/21/	Source: Biomass generation needs policy supply due to the high investment cost http://www.in-en.com/newenergy/html/newenergy-1754175414725040.html
/22/	http://energy.people.com.cn/GB/13161657.html
/23/	http://app.chinamining.com.cn/focus/EarthDay2009/2009-04-14/1239678253d23958.html
/24/	China Electric Power Yearbook 2004-2009
/25/	Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects
/26/	China Wind Power Report 2008 published by China Environmental Science Press in Sep. 2008
/27/	Chinese National Development and Reform Commission, Separate Power Plants from Network and Compete in Price to Enter Network, April 11, 2002, http://www.ndrc.gov.cn/xwfb/t20050708_28096.htm.
/28/	Notice on tariff management of wind power projects (Jijichu [2002] No. 2692) issued by National Development and Plan Committee on 10/12/2002
/29/	Tentative Measures for the Administration of Renewable Energy Power Price and Cost-sharing issued by NDRC on 04/01/2006 (Code: Fa Gai Jia Ge [2006] No.7) http://www.gov.cn/ztzl/2006-01/20/content_165910.htm
/30/	Notice on consolidating the Feed-in Tariff of Wind Power Project (Code: Fa Gai Jia Ge [2008] 1876) issued on 23/07/2008
/31/	Notice on adjustment of tariff in NECPG (Code: Fa Gai Jia Ge [2008] 1678) issued on 29/06/2008
/32/	Information Note on the Highest Tariffs Applied by the EB in Its Decision on Registration of Projects in the People's Republic of China (Version 2) published by EB
/33/	Notice of Clarifying Enforcement Time of Adjustment of Residual Value Ratio of Enterprise Fixed Assets issued by State Administration of Taxation of China
/34/	The Codes on Compiling Feasibility Study Report of Wind Farms issued by National Development Reform Committee (NDRC).
/35/	"Economic Evaluation Method and Parameters for Project Construction" (version 3)
/36/	"Wind Energy – the Facts" implemented by European Wind Energy Association (EWEA) published in Mar. 2009
/37/	IRR calculation spreadsheet of the Project
/38/	Bank Loan Contract on 28/03/2009
/39/	Commercial interest rates in China http://www.pbc.gov.cn/publish/zhengcehuobisi/631/2011/201107081425547994845

	98/20110708142554799484598_.html
/40/	The Document of the State Administration of Taxation, Decree No. 70 of Guoshuifa [2003] on 30 June 2003
/41/	Statute of Income Tax Law of China issued on 06/12/2007
/42/	Notice of Value added Tax Policy Regarding Products Using Certain Synthesized Resources and Other Products [2001] No. 198 issued by the Ministry of Finance and the State Administration of Taxation in Dec. 2001
/43/	Income Tax Law of China (Zhu Xi Ling No. 63) effective on 01/01/2008
/44/	IRR calculation spreadsheet
/45/	Notice of Value added Tax Policy Regarding Products Using Certain Synthesized Resources (Caishui [2008] No. 156) released by State Administration of Taxation on Dec 9th, 2008
/46/	Notice about National Value-Added Tax Reform and Transition (Caishui [2008] No. 170) released by State Administration of Taxation on 19/12/2008
/47/	http://finance1.jrj.com.cn/news/2008-03-28/000003465715.html
/48/	Statistics of wind power installed capacity in China (2007, 2008, 2009)
/49/	http://www.newenergy.org.cn/Html/9991/19991799.html
/50/	http://www.chinapower.com.cn/newsarticle/1005/new1005504.asp
/51/	Evidence of 40 pieces of stakeholder survey questionnaires
/52/	Statement of Modalities of Communication signed between Hegang Longyuan Wind Power Co., Ltd. and the buyer dated 21/11/2011
/53/	Power Purchase Agreement of the Project
/54/	Final Accounting Report of the Project issued by Zhongrui Yuehua Certified Public Accoutants

Category 2 Documents:
Background documents related to the design and/or methodologies employed in the design or other reference documents.

/Ref-1/	Validation and Verification Manual Version 01.2 dated 30/07/2010
/Ref-2/	ACM0002 version 12.1.0 dated 17/09/2010
/Ref-3/	Tool to calculate the emission factor for an electricity system Version 02.2.1 dated 29/09/2011
/Ref-4/	Tool for demonstration and assessment of additionality Version 05.2.1 dated 26/08/2008
/Ref-5/	Guidelines on the demonstration and assessment of prior consideration of the CDM Version 04 (Annex 13, EB 62)

/Ref-6/	Glossary of CDM terms Version 05
/Ref-7/	Guidelines for the Reporting and Validation of Plant Load Factors version 01 (EB48, Annex11)
/Ref-8/	Guidelines on the assessment of investment analysis version 05 (EB62 Annex 5)

Persons interviewed:
List persons interviewed during the validation or persons that contributed with other information that are not included in the documents listed above.

1	Mr. Ren Lei, Project Manager, Longyuan (Beijing) Carbon Asset Management Technology Co. Ltd.
2	Ms. Gao Xiaoqi, Project Manager, Longyuan (Beijing) Carbon Asset Management Technology Co. Ltd.
3	Mr. Ma Yifeng, Manager, Hegang Longyuan Wind Power Co., Ltd.
4	Mr. Xia Tongwei, Director, Company Service Center of Luobei County
5	Mr. Jiang Bin, Vice Director, DRC of Luobei County
6	Mr. Yuan Bo, Vice Director, Environment Protection Bureau of Luobei County
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9	Mr. Yang Hongjun, Resident, Damahe Forest Farm, Luobei County
10	Mr. Bai Changyong, Resident, Damahe Forest Farm, Luobei County

7. CURRICULA VITAE OF THE DOE'S VALIDATION TEAM MEMBERS

Ms. Li Yiting	Bureau Veritas Certification, China	Team Leader, Climate Change Lead Verifier She holds a Master Degree in Environmental Science. Before joining BV in 2009, she gained two and a half years of CDM technical working experience in P.R China. She obtained the certificate of CDM Lead Verifier, Lead Auditor for ISO 14001 and ISO 14064.

Mr. Wang Zhenning	Bureau Veritas Certification, China	Internal Technical Reviewer, Climate Change Lead Verifier He holds an MSc Degree in Environmental Technology and Bachelor Degree in Environmental Engineering. Before joining BV in 2010, he gained 4 years of technical experiences in the CDM industry in P.R China. He obtained the certificate of CDM Verifier in Nov 2010. He obtained the certificate of CDM Verifier and Lead Auditor for EMS ISO 14001.

APPENDIX A: COMPANY CDM PROJECT VALIDATION PROTOCOL

Table 1 Validation requirements based on the Validation and Verification Manual Version 01.2 (EB55 Annex 1)

CHECKLIST QUESTION	Ref.	§	COMMENTS		Draft Concl	Final Concl
1. Approval	VVM	44-50	COUNTRY A (*China*)	COUNTRY B (UK)		
a. Has the DNA of each Party indicated as being involved in the proposed CDM project activity in section A.3 of the PDD provided a written letter of approval? (If yes, provide the reference of the letter of approval, any supporting documentation, and specify if the letter was received from the project participant or directly from the DNA)	VVM	45	~~CAR-1:LoA from China should be provided.~~ The DNA of China has issued LoA for the Project in Jan. 2010 (No. 2342).	~~CAR-2:LoA from the Annex I Country should be provided.~~ The LoA from UK DNA dated 04/10/2011 has been provided and checked, its code is EA/TotalG&PLtd/03/2011, hence CAR 1 is closed.	~~CAR-1:~~ ~~CAR-2:~~	OK
b. Does the letter of approval from DNA of each Party confirm that : - The Party is a Party of the Kyoto Protocol - The participation is voluntary - In the case of the host Party, the proposed CDM project activity contributes to the sustainable development of the country - Refers to the precise proposed CDM project activity title in the PDD being submitted for registration	VVM	45	~~Pending on CAR-1:~~ Yes, the LoA from DNA of China confirm that: - The Party is a Party of the Kyoto Protocol - The participation is voluntary - the proposed CDM project activity contributes to the sustainable	~~Pending on CAR-2:~~ Yes, the LoA from DNA of UK confirm that: - The Party is a Party of the Kyoto Protocol - The participation is voluntary The LOA refers to the precise title of the Project.	~~Pending~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS		Draft Concl	Final Concl
			development of the country - The LOA refers to the precise title of the Project.			
c. Is(are) the letter(s) of approval unconditional with respect to (b) above?	VVM	46	No. It is unconditional with respect to (1.2) above.	No. It is unconditional with respect to (1.2) above.	OK	OK
d. Has(ve) the letter(s) of approval been issued by the respective Party's designated national authority (DNA) and is valid for the CDM project activity under validation?	VVM	47	~~Pending on CAR-1:~~ China's DNA is NDRC and the LoA is valid for the Project.	~~Pending on CAR-2:~~ UK DNA is FOEN and the LoA is valid for the Project.	~~Pending~~	OK
e. Is there doubt with respect to the authenticity of the letter of approval?	VVM	48	~~Pending on CAR-1:~~ No.	~~Pending on CAR-2:~~ No.	~~Pending~~	OK
f. If yes, was verified with the DNA that the letter of approval is authentic?	VVM	48	~~Pending on CAR-1:~~ N/A.	~~Pending on CAR-2:~~ N/A.	~~Pending~~	OK
2. Participation	VVM	51-54	*PP1 (Hegang Longyuan Wind Power Co., Ltd.)*	*PP2 (Total Gas & Power Ltd)*		
a. Are the project participants listed in tabular form in section A.3 of the PDD?	VVM	52	~~Pending on CAR-1:~~ Yes, all project participants have been listed in a consistent manner.	Yes.	~~Pending~~	OK
b. Does the DOE have a contractual relationship with the project participants?	EB50	Ann 48	Yes	N/A.	OK	OK
c. Is the information in section A.3 consistent with	VVM	52	~~Pending on CAR-1:~~	Yes	~~Pendin~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS		Draft Concl	Final Concl
the contact details provided in annex 1 of the PDD?			Yes		~~g~~	
d. Has the participation of each of the project participants been approved by at least one Party involved, either in a letter of approval or in a separate letter specifically to approve participation? (Provide reference of the approval document for each of the project participants)	VVM	52	~~Pending on CAR-1:~~ Yes, the Hegang Longyuan Wind Power Co., Ltd. has been approved by China DNA in a letter of approval (No. 2342).	~~Pending on CAR-2:~~ Yes, Total Gas & Power Ltd has been approved by UK DNA in a letter of approval (No. EA/TotalG&PLtd/03/2011).	~~Pending~~	OK
e. Are any entities other than those approved as project participants included in these sections of the PDD?	VVM	52	No.		OK	OK
f. Has the approval of participation issued from the relevant DNA?	VVM	53	~~Pending on CAR-1:~~ Yes, the approval of participation has been issued from China DNA – NDRC.	~~Pending on CAR-2:~~ Yes, the approval of participation has been issued from UK DNA – FOEN.	~~Pending~~	OK
g. Is there doubt with respect to (f) above?	VVM	53	~~Pending on CAR-1:~~ No, there is no doubt with respect to (f) above.	~~Pending on CAR-2:~~ No, there is no doubt with respect to (f) above.	~~Pending~~	OK
h. If yes, was verified with the DNA that the approval of participation is valid for the proposed CDM project participant?	VVM	53	N/A.	N/A.	OK	OK
3. Project design document	**VVM**	**55-57**				
a. Is the PDD in accordance with the applicable CDM requirements for completing the PDD?	VVM	56	Yes		OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
b. In CDM-PDD section A.1 are the following provided?	EB 41	Ann 12			
i. Title of project	EB 41	Ann 12	Heilongjiang Wangyunfeng Wind Power Project	OK	OK
ii. Current version number and date of document	EB 41	Ann 12	GSP Version number: 01, dated 26/08/2009 Final Version number: 02, dated 01/08/2011	OK	OK
c. In CDM-PDD section A.2 are following provided?	EB 41	Ann 12			
i. A brief description ot the project activity covering purpose which includes the scenario existing prior to the start or project, present scenario and baseline scenario	EB 41	Ann 12	Yes. The purpose of the proposed project is to generate electricity from wind resources to deliver the electricity to the Northeast China Power Grid (NECPG). For the proposed project, The scenario existing prior to the start of the implementation of the Project is to provide the same annual electricity output as the Project by NECPG. The baseline scenario is the same as the scenario existing prior to the start of implementation of the project activity.	OK	OK
ii. Explanation on how the GHG emission reductions are effected	EB 41	Ann 12	Yes. The Project involves the installation of 58 turbines, each of which has a rated output of 850kW, providing a total capacity of 49.3MW. The annual grid-connected output of the Project is estimated to be 120,069 MWh. The Project activity will achieve greenhouse gas (GHG) emission reductions by avoiding CO_2 emissions from the	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			business-as-usual scenario electricity generated by those fossil fuel-fired power plants connected into NECPG. It is estimated that annual emission reductions are 123,430tCO_2e.		
iii. The PP's views on the contribution of project activity to sustainable development	EB 41	Ann 12	Yes. The proposed project will contribute to the sustainable development of the host country through the following aspects: ➢ To contribute to local economy development by providing electricity to meet local increasing energy demands; ➢ To reduce GHG emissions and to mitigate the emissions of other pollutants caused from local coal-fired power plants compared with a business-as-usual scenario by displacing part of electricity from fossil fuel-fired power plants; ➢ To be in accordance with the development priority of China energy industry, and to help diversify energy mix of NECPG by increasing the share of renewable energy; ➢ To create plenty of short-term employment opportunities and permanent jobs for the local people during the construction and operation period of the Project.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
iv. Are there any changes/modifications compared to the webhosted PDD?	EB 41	Ann 12	There are no changes/modifications compared to the webhosted PDD.	OK	OK
d. In CDM-PDD section A.3 are following provided in the tabular format?	EB 41	Ann 12			
i. List of project participants and parties	EB 41	Ann 12	Yes. The PP has been provided.	OK	OK
ii. Identification of Host Party	EB 41	Ann 12	Yes. China is Host Party.	OK	OK
iii. Indication whether the Party wishes to be considered as project participant	EB 41	Ann 12	Yes. Both Parties do not wish to be PP.	OK	OK
e. In CDM-PDD section A.4.1 are following provided?	EB 41	Ann 12			
i. Technical description, location, host party(ies) and address as required	EB 41	Ann 12	Yes. The Project is located in the Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, P.R. of China.	OK	OK
ii. Detailed physical location with unique identification of the project activity (eg. Longitude/latitude)	EB 41	Ann 12	Yes. The geographical coordinates are longitude 130°39′56″ East and latitude 47°54′34‴ North.	OK	OK
iii. Are there any changes/modifications compared to the webhosted PDD?	EB 41	Ann 12	No. There are no changes/modifications compared to the webhosted PDD.	OK	OK
f. In CDM-PDD section A.4.2 is the list of categories of project activities provided?	EB 41	Ann 12	Scope 1: Energy Industries (renewable sources)	OK	OK
g. In CDM-PDD section A.4.3 are following provided?	EB 41	Ann 12			
i. A description of how environmentally safe and	EB	Ann	Yes. The technologies employed in the proposed	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
sound technology, and know-how, is transferred to the Host Party(ies)	41	12	project activity are advanced domestic technologies, which is no technology transfer activity involved.		
ii. Explanation of purpose of project activity with scenario existing prior to the start of project, scope or present activities and the baseline scenario	EB 41	Ann 12	Yes. The purpose of the Project is to generate wind power and deliver it to NECPG. The baseline scenario is the same as the scenario existing prior to the start of implementation of the project activity.	OK	OK
iii. List and arrangement of the main manufacturing/production technologies, systems and equipments involved	EB 41	Ann 12	~~CAR-3:As per the latest CDM PDD guideline, the lifetime of the equipments based on manufacturer's specifications and load factor should be listed in section A.4.3 of PDD. In accordance with the FSR, operational period of the Project is 20 years. A correction in Section B.5 and C.1.2 should be provided.~~ After checking the revised PDD, it was found that relevant information was correctly included in section A.4.3.	~~CAR-3:~~	OK
iv. The emissions sources and GHGs involved	EB 41	Ann 12	~~CL-1: The emissions sources and GHGs involved should be clarified in section A.4.3~~To reduce greenhouse gas emissions of CO_2 produced in NECPG.	~~CL-1:~~	OK
v. Are there any changes/modifications compared to the webhosted PDD?	EB 41	Ann 12	No. There are no changes/modifications compared to the webhosted PDD.	OK	OK
h. In CDM-PDD section A.4.4 is the estimation of emission reductions provided as requested in a	EB 41	Ann 12	7×3 renewable crediting periods chosen.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
tabular format?			Annual emission reductions of 123,430 tCO_2e are estimated for the first crediting period.		
i. In CDM-PDD section A.4.5 is Information regarding Public funding provided?	EB 41	Ann 12	Yes. No public founding involved confirmed with the approved FSR.	OK	OK
j. In CDM-PDD section B.1 are following provided?	EB 41	Ann 12			
i. The approved methodology and version number	EB 41	Ann 12	Version 10 of ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”. ~~CL-2:ACM0002 version 12.1.0 was valid from 26/11/2010. The applied version of methodology should be updated.~~~~*CAR-4:The latest “Tool to calculate the emission factor for an electricity system” version 02.2.1 should be used.*~~ By checking the PDD, methodology ACM0002 version 12.1.0, Tool to calculate the emission factor for an electricity system version 02.2.1 has been used in the updated PDD.	~~CL-2:CAR-4:~~	OK
ii. Any methodologies or tools which the above approved methodology draws upon and their version number	EB 41	Ann 12	~~Pending on CL-2:and CAR-4:.~~ “Tool for demonstration and assessment of additionality” version 05.2.1 and “tool to calculate the emission factor for an electricity system” version 02.2.1 was used in the updated	~~Pending~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			PDD.		
k. In CDM-PDD section B.2 are following provided?	EB 41	Ann 12			
i. Justification to the choice of methodology that the project activity meets each of the applicability conditions	EB 41	Ann 12	~~Pending on close CL-2:.~~ The Project is a new wind power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (greenfield plants). The Project does not involve switching from fossil fuels to renewable energy at the site of the project activity.	~~Pending~~	OK
ii. Documentations with references that had been used. This can be provided in Annex 3 instead	EB 41	Ann 12	~~Pending on close CL-2: and CAR-4:.~~ ACM0002 ver. 12.1.0 *"Consolidated methodology for grid-connected electricity generation from renewable sources"* "Tool for demonstration and assessment of additionality" version 05.2.1 "Tool to calculate the emission factor for an electricity system" version 02.2.1	~~Pending~~	OK
l. In CDM-PDD section B.3 are following provided?	EB 41	Ann 12			
i. Description of all sources and gases included in the project boundary in the table	EB 41	Ann 12	Yes. The description as per the methodology is provided in a table and a flow diagram of the proposed project in the PDD section B.3.	OK	OK
ii. A flow diagram of the project boundary physically delineating the project activity	EB 41	Ann 12	Yes.	OK	OK
iii. The flow diagram with all equipments, systems and flows of mass and energy etc	EB 41	Ann 12	Yes.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
m. In CDM-PDD section B.4 are following provided?	EB 41	Ann 12			
i. Explanation how the most plausible baseline scenario is identified in accordance with the selected baseline methodology	EB 41	Ann 12	~~CL-3:The baseline scenario has been identified directly in ACM0002. Please explain the baseline scenario identification in accordance with ACM0002 version 12.1.0 in section B.4. of the PDD.~~ The revised description is consistent with the applied methodology.	~~CL-3:~~	OK
ii. Justification of key assumptions and rationales	EB 41	Ann 12	Not applicable.	OK	OK
iii. Transparent illustration of all data used to determine the baseline scenario (variables, parameters, data sources, etc.)	EB 41	Ann 12	Not applicable.	OK	OK
iv. A transparent and detailed description of the identified baseline scenario, including a description of the technology that would be employed and/or the activities that would take place in the absence of the proposed project activity	EB 41	Ann 12	~~Pending on CL-3:~~ Methodology ACM0002 prescribes the baseline scenario and no further analysis required, thus there is no need to take steps to identify the baseline scenarios.	~~Pending~~	OK
v. Are there any changes/modifications compared to the webhosted PDD?	EB 41	Ann 12	~~Pending on CL-3:~~ There are no changes/modifications to the identified baseline scenario compared to the webhosted PDD.	~~Pending~~	OK
n. In CDM-PDD section B.5 are following provided?	EB 41	Ann 12			
i. Explanation of how and why this project activity	EB	Ann	Yes.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
is additional and therefore not the baseline scenario in accordance with the selected baseline methodology	41	12	The Tool for the Demonstration and Assessment of Additionality version 05.2.1 was followed to demonstrate its additionality.		
ii. Justification of key assumptions and rationales	EB 41	Ann 12	Yes.	OK	OK
iii. Transparent illustration of all data used to determine the baseline scenario (variables, parameters, data sources etc)	EB 41	Ann 12	Yes.	OK	OK
iv. Evidence that the incentive from the CDM was seriously considered in the decision to proceed with the project activity, if the starting date of the project activity is before the date of validation	EB 41	Ann 12	Yes. The starting date of the Project is 12/05/2009. The timeline and evidences are verified, and it is confirmed the identified starting date is the earlest date at which the implementation or construction or real action of the Project began.	OK	OK
o. In CDM-PDD section B.6.1 are following provided?	EB 41	Ann 12			
i. Explanation as to how the procedures, in the approved methodology to calculate project emissions, baseline emissions, leakage emissions and emission reductions are applied to the proposed project activity	EB 41	Ann 12	Complying with ACM0002, the "*Tool to calculate the emission factor for an electricity system*" ver. 1.1 is used in the GSP PDD. ~~Pending on CAR-4:.~~ "*Tool to calculate the emission factor for an electricity system*" ver. 02.2.1 is used in the revised PDD.	~~Pending~~	OK
ii. Equations used in calculating emission redutions	EB 41	Ann 12	~~Pending on CAR-4:.~~	~~Pending~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Yes. The equations are in accordance with "*Tool to calculate the emission factor for an electricity system*" ver. 02.2.1 in the revised PDD.		
iii. Explanation and justification for all relevant methodological choices, including different scenarios or cases, options and default values	EB 41	Ann 12	~~Pending on CAR-4:.~~ Yes. The explanation and justification are in accordance with "*Tool to calculate the emission factor for an electricity system*" ver. 02.2.1 in the revised PDD.	~~Pending~~	OK
p. In CDM-PDD section B.6.2 are following provided?	EB 41	Ann 12			
i. A compilation of information on the data and parameters that are not monitored throughout the crediting period but that are determined only once and thus remains fixed throughout the crediting period and that are available when validation is undertaken	EB 41	Ann 12	Yes.	OK	OK
ii. The actual value applied	EB 41	Ann 12	Yes. The official values before validation are applied.	OK	OK
iii. Explanation and justification for the choice of the source of data	EB 41	Ann 12	Yes. The latest official values before validation are applied.	OK	OK
iv. Clear and transparent references or additional documentation in Annex 3	EB 41	Ann 12	Yes. The values and references have been provided in Annex 3.	OK	OK
v. Where values have been measured, a description of the measurement methods and procedures (e.g. which standards have been	EB 41	Ann 12	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
used), indicated the responsible person/entity having undertaken the measurement, the date of measurement(s) and the measurement results					
q. In CDM-PDD section B.6.3 are following provided?	EB 41	Ann 12			
i. A transparent *ex ante* calculation of project emissions, baseline emissions (or, where applicable, direct calculation of emission reductions) and leakage emissions expected during the crediting period, applying all relevant equations provided in the approved methodology	EB 41	Ann 12	~~Pending on CAR-4:~~ The revised calculation process is in line with the steps taken prescribed in “Tool-Grid EF” and addressed in PDD B.6.3 and Annex 3.	~~Pending~~	OK
ii. Documentation how each equation is applied, in a manner that enables the reader to reproduce the calculation	EB 41	Ann 12	Yes. The spreadsheets are used.	OK	OK
iii. Additional background information and or data in Annex 3, including relevant electronic files (i.e. spreadsheets)	EB 41	Ann 12	Yes. The calculation spreadsheet has been presented for re-produce.	OK	OK
r. In CDM-PDD section B.6.4 are the results of the *ex ante* estimation of emission reductions for all years of the crediting period, provided in a tabular format?	EB 41	Ann 12	Yes.	OK	OK
s. In CDM-PDD section B.7.1 are following provided?	EB 41	Ann 12			
i. Specific information on how the data and parameters that need to be monitored would	EB 41	Ann 12	*EG y* –net electricity supplied to NECPG	~~CL-4:~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
actually be collected during monitoring for the project activity			~~CL-4: The monitoring information about how the electricity exported and imported will be monitored should be provided in section B.7.1 of the PDD.~~ $EG_{export,y}$, $EG_{import,y}$ and $EG_{facility,y}$ are listed in B.7.1.		
ii. For each parameter the following below information, using the table provided:	EB 41	Ann 12			
a. The source(s) of data that will be actually used for the proposed project activity (e.g. which exact national statistics). Where several sources may be used, explain and justify which data sources should be preferred.	EB 41	Ann 12	Yes, the electricity will be measured by meters in the substation.	OK	OK
b. Where data or parameters are supposed to be measured, specify the measurement methods and procedures, including a specification which accepted industry standards or national or international standards will be applied, which measurement equipment is used, how the measurement is undertaken, which calibration procedures are applied, what is the accuracy of the measurement method, who is the responsible person/entity that should undertake the measurements and what is the measurement interval; (i) A description of the QA/QC procedures (if any) that should be applied; (ii) Where	EB 41	Ann 12	~~Pending on CL-4:.~~ The accuracy of meters will be no less than 0.5%, which is included in the PDD. The electricity will be continuously measured and recorded monthly. Data may be verified against the Receipt of sales. The meters are expected to be calibrated annually.	~~Pending~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
relevant: any further comment. Provide any relevant further background documentation in Annex 4.					
t. In CDM-PDD section B.7.2 are following provided?	EB 41	Ann 12			
i. A detailed description of the monitoring plan	EB 41	Ann 12	~~Pending on CL-4:.~~ ~~*CL-5: Please clarify calibrating frequency of the monitoring meter and the emergency measures that will be taken when the meters cannot work normally in section B.7.2 of the PDD.*~~ The metering equipments at the substation are calibrated and tested yearly and the emergency measures have been provided, so this CL is closed out. The revised MP is considered complete and contains a detailed description of monitoring plan.	~~CL-5:~~	OK
ii. The operational and management structure that the project operator will implement in order to monitor emission reductions and any leakage effects generated by the project activity	EB 41	Ann 12	~~Pending on CL-4: and CL-5:.~~ The revised MP is considered complete.	~~Pending~~	OK
iii. The responsibilities for and institutional arrangements for data collection and archiving	EB 41	Ann 12	Yes. The PP will take the responsibility.	OK	OK
iv. Indication that the monitoring plan reflect good monitoring practice appropriate to the type of project activity	EB 41	Ann 12	~~Pending on CL-4: and CL-5:.~~ The revised MP is considered complete, and the monitoring plan can reflect good monitoring practice appropriate to the type of the Project. .	~~Pending~~	OK
v. Relevant further background information in Annex 4	EB 41	Ann 12	There is no further background information.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
u. In CDM-PDD section B.8 are following provided?	EB 41	Ann 12			
i. Date of completion of the application of the methodology to the project activity study in DD/MM/YYYY	EB 41	Ann 12	Yes. The date is 26/08/2009 in PDD version 01, and 01/08/2011 in PDD version 02.	OK	OK
ii. Contact information of the person(s)/entity(ies) responsible for the application of the baseline and monitoring methodology to the project activity	EB 41	Ann 12	Yes. Contact information of Longyuan (Beijing) Carbon Asset Management Technology Co. Ltd. is lised.	OK	OK
iii. Indication if the person/entity is also a project participant listed in Annex 1	EB 41	Ann 12	Yes. All the persons listed are not project participants.	OK	OK
v. In CDM-PDD section C.1.1, Is the project's starting date clearly defined and evidenced?	EB 41	Ann 12	Yes. Relevant evidence related to the identification of start date of the Project has been checked and found that 12/05/2009, the signed date of turbine purchase contract, was identified as the start date of the Project, which complied with the latest CDM glossary.	OK	OK
w. In CDM-PDD section C.1.2 is the expected operational lifetime of the project activity in years and months provided?	EB 41	Ann 12	Yes.	OK	OK
x. In CDM-PDD section C.2, is the assumed crediting time clearly defined and reasonable (renewable crediting period of max. three x 7 years or fixed crediting period of max. 10 years)? ?	EB 41	Ann 12	Yes. Renewable crediting period of max. three x 7 years has been defined. 01/03/2012.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
y. In CDM-PDD section D, are the conclusions and all references to support documentation of an environmental impact assessment undertaken in accordance with the procedures as required by the Host Party, if environmental impacts are considered significant by the project participants or the Host, provided?	EB 41	Ann 12	Yes. According to the results of EIA and the approval from the Environmental Protection Bureau, the impacts on the environment are not significant.	OK	OK
z. In CDM-PDD section E.1 are the following provided?	EB 41	Ann 12			
i. The process by which comments by local stakeholders have been invited and compiled. An invitation for comments by local stakeholders shall be made in an open and transparent manner, in a way that facilities comments to be received from local stakeholders and allows for a reasonable time for comments to be submitted.	EB 41	Ann 12	Yes. In May 2009, staff from the project owner carried out a survey of local residents possibly be impacted in the area where the Project is sited to collect public comments and attitudes towards the Project before the construction starting. The comments on the project activity by the local stakeholders have been invited and complied both in a symposium and questionnaires.	OK	OK
ii. The project activity is described in a manner, which allows the local stakeholders to understand the project activity, taking into account confidentiality provisions of the CDM modalities and procedures.	EB 41	Ann 12	Yes. A detailed description of the proposed project is provided in the questionnaires distributed to collect the stakeholders' comments.	OK	OK
iii. The local stakeholder process has been completed before submitting the proposed project activity to the DOE for validation.	EB 41	Ann 12	Yes, the local stakeholder process has been conducted on 06/05/2009, which is before submitting the Project to the DOE for validation.	OK	OK
aa. In CDM-PDD section E.2 are following provided?	EB	Ann			

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
	41	12			
i. Identification of local stakeholders that have made comments	EB 41	Ann 12	Yes. The survey was made among the potential stakeholders, mostly including local residents and local government.	OK	OK
ii. A summary of this comments.	EB 41	Ann 12	~~*CL-6: The summary of the comments received is not complete.*~~ The local residents supported the Project, and the project owner has an overall environment-friendly plan to guarantee the project has the minimum negative impact on the environment during the project construction and operation.	~~CL-6:~~	OK
bb. In CDM-PDD section E.3 is the explanation of how due account have been taken of comments received from local stakeholders provided?	EB 41	Ann 12	~~Pending on CL-6:~~ No negative comments have been received on the Project. There has therefore been no need to modify the project due to comments received.	~~Pending~~	OK
cc. In CDM-PDD Annex 1 are the following provided?	EB 41	Ann 12			
i. Contact information of project participants	EB 41	Ann 12	Yes.	OK	OK
ii. For each organisation listed in section A.3 the following mandatory fields: Organization, Name of contact person, Street, City, Postfix/ZIP, Country, Telephone and Fax or e-mail	EB 41	Ann 12	Yes.	OK	OK
dd. In CDM-PDD Annex 2 is information from Parties included in Annex I on sources of public funding for the project activity which shall provide an affirmation that such funding does not result in a	EB 41	Ann 12	There is no public funding for the Project.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
diversion of official development assistance and is separate from and is not counted towards the financial obligations of those Parties provided?					
ee. In CDM-PDD Annex 3 is the background information used in the application of the baseline methodology provided?	EB 41	Ann 12	Yes.	OK	OK
ff. In CDM-PDD Annex 4 is the background information used in the application of the monitoring methodology provided?	EB 41	Ann 12	There is no additional information.	OK	OK
4. Project description	**VVM**	**58-64**			
a. Is the description of the proposed CDM project activity as contained in the PDD:	VVM	59			
i. sufficiently covering all relevant elements?	VVM	59	Yes.	OK	OK
ii. acurate?	VVM	59	Yes. The proposed project involves the installation of 58 turbines, each of which has a rated output of 850kW, providing a total capacity of 49.3MW. The annual grid-connected output of the proposed project is estimated to be 120,069 MWh. The Project activity will achieve greenhouse gas (GHG) emission reductions by avoiding CO_2 emissions from the business-as-usual scenario electricity generated by those fossil fuel-fired power plants connected into NECPG. It is estimated that annual emission reductions are 123,430tCO_2e.	OK	OK
iii. providing the reader with a clear understanding of the nature of the proposed CDM project	VVM	59	Yes.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
activity?					
iv. Are there any changes/modifications compared to the webhosted PDD?	VVM	59	No. There are no changes/modifications compared to the webhosted PDD.	OK	OK
b. Is the proposed CDM project activity in existing facilities or or utilizing existing equipments?	VVM	60	No.	OK	OK
c. Is the CDM project activity one of the following types:	VVM	60			
i. Large scale?	VVM	60	Yes.	OK	OK
ii. Non-bundled small scale projects with emission reductions exceeding 15,000 tonnes per year?	VVM	60	No.	OK	OK
iii. Bundled small scale projects, each with emission reductions not exceeding 15,000 tonnes?	VVM	60	No.	OK	OK
d. If yes to (b) or (c) above, was a physical site inspection conducted to confirm that the description in the PDD reflects the proposed CDM project activity, unless other means are specified in the methodology?	VVM	60	Yes. A physical site inspection was conducted by Ms. Li Yiting from Bureau Veritas Certification on 21/10/2009.	OK	OK
e. If yes to (c.iii) above, was the number of physical site visits base on samping?	VVM	60	N/A.	OK	OK
f. If yes is the sampling size appropriately justified through statistical analysis?	VVM	60	N/A.	OK	OK
g. For other individual proposed small scale CDM project activities with emission reductions not exceeding 15,000 tonnes per year, was a physical site inspection conducted?	VVM	61	N/A.	OK	OK
h. For all other proposed CDM project activities not	VVM	62	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
referred to in VVM paragraphs 59 – 61, was a physical site inspection conducted?					
i. If no, was it appropriately justified?	VVM	62	N/A.	OK	OK
j. Does the proposed CDM project activity involve the alteration of an existing installation or process?	VVM	63	N/A.	OK	OK
k. If yes, does the project description clearly state the differences resulting from the project activity compared to the pre-project situation?	VVM	63	N/A.	OK	OK
5. Baseline and monitoring methodology					
a. General requirement	**VVM**	**65-67**			
a. Do the baseline and monitoring methodologies selected by the project participants comply with the methodologies previously approved by the CDM Executive Board?	VVM	65	Yes. ACM0002 was applied.	OK	OK
b. Is the selected methodology applicable to the project activity?	VVM	66	Refer to (5.b.a) below	-	OK
c. Had the PP correctly applied the selected methodology?	VVM	66	Refer to (5.b.d) below	-	OK
d. Had the selected methodology been correctly applied with respect to project boundary?	VVM	67	Refer to (5.c) below	-	OK
e. Had the selected methodology been correctly applied with respect to baseline identification?	VVM	67	Refer to (5.d) below	-	OK
f. Had the selected methodology been correctly applied with respect to Algorithms and/or formulae used to determine emission reductions?	VVM	67	Refer to (5.e) below	-	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
g. Had the selected methodology been correctly applied with respect to additionality?	VVM	67	Yes. "Tool for demonstration and assessment of additionality" version 05.2.1 was used in the PDD.	OK	OK
h. Had the selected methodology been correctly applied with respect to monitoring methodology?	VVM	67	Yes. The monitoring methodology ACM 0002 'Consolidated baseline methodology for grid-connected electricity generation from renewable sources' was applied.	OK	OK
b. Applicability of the selected methodology to the project activity	VVM	68-77			
a. Is the methodology correctly quoted?	VVM	70	ACM0002 version 10 is used in the PDD version 01. ~~Pending on CL-2:.~~ ACM0002 version 12.1.0 is used in the final version PDD.	~~Pending~~	OK
b. Are the applicability conditions of the methodology ACM0002 Ver. 12.1.0 met?	VVM	71			
i. This methodology is applicable to grid-connected renewable power generation project activities that (a) install a new power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (greenfield plants); (b) involve a capacity addition; (c) involve a retrofit of (an) existing plant(s); or (d) involve a replacement of (an) existing plant(s).	ACM	0002	~~Pending on close CL-2:.~~ The Project is a new wind power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (greenfield plants).	~~Pending~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
ii. The project activity is the installation, capacity addition, retrofit or replacement of a power plant/unit of one of the following types: hydro power plant/unit (either with a run-of-river reservoir or an accumulation reservoir), wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit	ACM	0002	~~Pending on close CL-2:.~~ The Project is the installation of a wind power plant.	~~Pending~~	OK
iii. In the case of capacity additions, retrofits or replacements: the existing plant started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion or retrofit of the plant has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	ACM	0002	~~Pending on close CL-2:.~~ Not applicable.	~~Pending~~	OK
iv. In case of hydro power plants, one of the following conditions must apply: - The project activity is implemented in an existing reservoir, with no change in the volume of reservoir; or - The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the Project Emissions section, is	ACM	0002	Not applicable.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
greater than 4 W/m^2; or - The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the Project Emissions section, is greater than 4 W/m^2.					
v. The methodology is not applicable to the following conditions. Please confirm • Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity • Biomass fired power plants; • Hydro power plants that result in new reservoirs or in the increase in existing reservoirs where the power density of the power plant is less than 4 W/m2.	ACM	0002	The Project does not involve switching from fossil fuels to renewable energy at the site of the project activity.	OK	OK
c. Is the project activity expected to result in emissions other than those allowed by the methodology?	VVM	71	No.	OK	OK
d. Is the choice of the methodology justified?	VVM	71	Yes. ACM0002 has been chosen.	OK	OK
e. Have the project participants shown that the project activity meets each of the applicability conditions of the approved methodology?	VVM	71	Refer to (5.b.d) above	-	OK
f. Have the project participants shown that the project activity meets each of the applicability conditions of any tool or other methodology component referred to the methodology?	VVM	71	Yes. The Project meets the applicability conditions of tool to calculate the emission factor for an electricity system and tool for demonstration and assessment of additionality.	OK	OK
g. Is the DOE, based on local and sectoral	VVM	71	Yes.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
knowledge, aware that comparable information is available from sources other than that used in the PDD?					
h. If yes, was the PDD cross checked against the other sources to confirm that the project activity meets the applicability conditions of the methodology? (provide the reference to these choices)	VVM	71	Yes. BVC has checked FSR approval, EIA approval, equipment purchase contracts and conducted on-site observation and confirmed the Project meets the applicability of the applied methodology.	OK	OK
i. Can a determination regarding the applicability of the selected methodology to the proposed CDM project activity be made?	VVM	72	Yes.	OK	OK
j. If no, clarification of the methodoloy was requested, in accordance with the guidance provided by the CDM Executive Board?	VVM	72	N/A.	OK	OK
k. If answer to (5.b.b) above is “no”, revision or deviation from the methodology was requested, in accordance with the guidance provided by the CDM Executive Board?	VVM	73	N/A.	OK	OK
l. If yes to (5.b.j) and (5.b.k) above, a request for registration was submited before the CDM Executive Board has approved the proposed deviation or revision?	VVM	74	N/A.	OK	OK
c. Project boundary	VVM	78-80			
a. Is the delineation in the PDD of the project boundary correct and include identification of all locations, processes and equipment including secondary equipment and associated processes such as logistics etc.?	VVM	79	Yes. The spatial extent of the project boundary includes the project power plant and all the power plants connected physically to Northeast China Grid that the Project is connected to. Electricity generated	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			by the project will be delivered to Northeast China Grid. According to the China' Regional Grid Baseline Emission Factors issued by China's DNA which provides the delineation of grid boundaries, Northeast China Grid is the grid boundary of the project.		
b. Does the delineation in the PDD of the project boundary meet the requirements of the selected baseline?	VVM	79	Yes. The defined project boundary is in line with ACM0002 ver 12.1.0. and all emission sources and GHGs have been included in the project boundary.	OK	OK
c. Have changes been made to the project boundary in comparison to the webhosted PDD? If yes please comment on the reason for the changes.	VVM	79	No.	OK	OK
d. Have all sources and GHGs required by the methodology been included within the project boundary?	VVM	79	Yes.	OK	OK
e. Does the methodology allow project participant to choose whether a source or gas is to be included within the project boundary	VVM	79	No.	OK	OK
f. If yes, have the project participants justified that choice?	VVM	79	N/A.	OK	OK
g. If yes, is the justification provided reasonable? (provide reference to the supporting documented evidence provided by the project participants)	VVM	79	N/A.	OK	OK
d. Baseline identification	VVM	81-88			

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
a. Does the methodology require several alternative scenarios to be considered in the identification of the most reasonable baseline scenario?	VVM	83	No. The ACM0002 prescribe the baseline scenario.	OK	OK
b. If yes, are all scenarios that are considered by the project participants and are supplementary to those required by the methodology reasonable in the context of the proposed CDM project activity?	VVM	83	N/A.	OK	OK
c. Has any reasonable alternative scenario been excluded?	VVM	83	N/A.	OK	OK
d. Is the baseline scenario identified reasonably supported by:	VVM	84			
i. Assumptions?	VVM	84	N/A.	OK	OK
ii. Calculations?	VVM	84	N/A.	OK	OK
iii. Rationales?	VVM	84	N/A.	OK	OK
e. Are the documents and sources referred to in the PDD correctly quoted and interpreted?	VVM	84	Yes. ACM0002 and Chinese official EF	OK	OK
f. Was the information provided in the PDD cross checked with other verifiable and credible sources, such as local expert opinion, if available? (idendify the sources)	VVM	84	Yes. The information has been cross checked with the applied methodology and valid official EF data at the time of validation and found consistent.	OK	OK
g. Have all applicable CDM requirements been taken into account in the identification of the baseline scenario for the proposed CDM project activity?	VVM	85	Yes. The ACM0002 prescribe the baseline scenario.	OK	OK
h. Have all relevant policies and circumstances been identified and correctly considered in the PDD, in accordance with the guidance by the CDM Executive Board?	VVM	85	Yes. The ACM0002 prescribe the baseline scenario.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
i. Does the PDD provide a verifiable description of the identified baseline scenario, including a description of the technology that would be employed and/or the activities that would take place in the absence of the proposed CDM project activity?	VVM	86	Yes. The baseline scenario of the proposed project can be identified as the electricity delivered to the grid by the proposed project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources within the NECPG, as reflected in the combined margin (CM) calculated.	OK	OK
e. Algorithms and/or formulae used to determine emission reductions	VVM	89-93			
a. Have the equations and parameters in the PDD been correctly applied with respect those in the select approved methodology?	VVM	90	~~Pending on CL-2:.~~ Yes. The revised steps and equations are complying with the selected methodology.	~~Pending~~	OK
b. Does the methodology provide for selection between different options for equations or parameters?	VVM	90	Yes.	OK	OK
c. If yes, has adequate justification been provided (based on the choice of the baseline scenario, context of the proposed CDM project activity and other evidence provided)?	VVM	90	~~*CAR-5:It is stated in section B.6.1. of the PDD that there were electricity imports from NWPG to NECPG. However, from the official statistic and the calculation issued on 02/07/2009 by Chinese DNA, there were no electricity imports from NWPG to NECPG. Please correct relevant description in the PDD.*~~ According to the official statistic and the calculation issued on 02/07/2009, the relevant sections have been revised and updated, so this CAR is closed out.	~~CAR-5:~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			~~Pending on CL-2:.~~ ACM0002 version 12.1.0 has been applied in the revised PDD, so ~~CL-2:~~ is closed out. By checking the revised PDD, adequate justification has been provided and correct equations and parameters been used in accordance with the methodology selected.		
d. If yes, have correct equations and parameters been used, in accordance with the methodology selected?	VVM	90	Refer to (5.e.b) above	-	OK
e. Will data and parameters be monitored throughout the crediting period of the proposed CDM project activity?	VVM	91	No.	OK	OK
f. If no, and these data and parameters will remain fixed throughout the crediting period, are all data sources and assumptions:	VVM	91			
i. Appropriate and correct?	VVM	91	~~Pending on close CAR-4: and CL-2:.~~ Yes, all data source used are appropriate and calculations are correct.	~~Pending~~	OK
ii. Applicable to the proposed CDM project activity?	VVM	91	~~Pending on close CAR-4: and CL-2:.~~ Yes, all equations and parameters used are applicable to the Project.	~~Pending~~	OK
iii. Resulting in a conservative estimate of the emission reductions?	VVM	91	~~Pending on close CAR-4: and CL-2:.~~ Yes, all equations and parameters will result in a conservative estimate of the emission reductions.	~~Pending~~	OK
g. Will data and parameters be monitored on implementation and hence become available only after validation of the project activity?	VVM	91	No.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
h. If yes, are the estimates provided in the PDD for these data and parameters reasonable?	VVM	91	N/A.	OK	OK
6. Additionality of a project activity	VVM	94-97			
a. Does the CDM-PDD state the latest version of the additionality tool being used?	VVM	95	Yes. The approved "Tool for the Demonstration and Assessment of Additionality" version 05.2.1 is used in the PDD.	OK	OK
b. Were the following steps of the tool to assess additionality used:	EB 39	Ann 10			
i. Identification of alternatives to the project activity?	EB 39	Ann 10	Yes.	OK	OK
ii. Investment analysis to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible?	EB 39	Ann 10	Yes.	OK	OK
iii. Barriers analysis?	EB 39	Ann 10	No.	OK	OK
iv. Common practice analysis?	EB 39	Ann 10	Yes.	OK	OK
c. In step 1 (i) have all the sub-steps as below been followed?	EB 39	Ann 10			
i. Sub-step 1a: Define alternatives to the project activity	EB 39	Ann 10	Yes.	OK	OK
ii. Sub-step 1b: Consistency with mandatory laws and regulations	EB 39	Ann 10	Yes.	OK	OK
d. Have the following alternatives been included while defining alternatives as per sub-step 1a?	EB 39	Ann 10			

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
i. (a) The proposed project activity undertaken without being registered as a CDM project activity;	EB 39	Ann 10	Yes. Alternative a) is identified: To implement the proposed project activity, but not as a CDM project activity.	OK	OK
ii. (b) Other realistic and credible alternative scenario(s) to the proposed CDM project activity scenario that deliver outputs services or services with comparable quality, properties and application areas, taking into account, where relevant, examples of scenarios identified in the underlying methodology;	EB 39	Ann 10	Yes. Alternative b) and c) are identified. Alternative b): To construct a thermal power plant with the same electricity output as the Project; Alternative c): To construct a power plant using other renewable resources with the same electricity output as the Project.	OK	OK
iii. (c) If applicable, continuation of the current situation (no project activity or other alternatives undertaken).	EB 39	Ann 10	Yes. Alternative d) is identified. Alternative d): To provide the same annual electricity output as the Project by NECPG.	OK	OK
e. Has the project participant included the technologies or practices that provide outputs or services with comparable quality, properties and application areas as the proposed CDM project activity and that have been implemented previously or are currently being introduced in the relevant country/region?	EB 39	Ann 10	No.	OK	OK
f. Has the outcome of Step 1a: Identified realistic and credible alternative scenario(s) to the project activity done correctly? Please briefly mention the outcome.	EB 39	Ann 10	~~CL-8:Please clearly clarify the outcome of Step 2 in section B.5. of the PDD according to the latest version of 'Tool *for The Demonstration and Assessment of Additionality*'.~~ Alternative c) was excluded due to technology development and high cost for power generation	~~CL-8:~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			of solar PV, geothermal and biomass projects. And there are no enough water resources in the area of project site.		
g. Is the alternative(s) in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution.?	EB 39	Ann 10	Yes. Alternative b) is not in compliance with the laws of China, and all the other options comply with current mandatory laws and regulations.	OK	OK
h. If an alternative does not comply with all mandatory applicable legislation and regulations, has it been shown that, based on an examination of current practice in the country or region in which the law or regulation applies, those applicable legal or regulatory requirements are systematically not enforced and that noncompliance with those requirements is widespread in the country?	EB 39	Ann 10	No.	OK	OK
i. Has the outcome of Step 1b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations done correctly? Please state the outcome.	EB 39	Ann 10	Yes. The proposed project activity is not the only alternative amongst the project participants that is in compliance with mandatory regulations with which there is general compliance.	OK	OK
j. Has PP selected Step 2 (Investment analysis) or Step 3 (Barrier analysis) or both Steps 2 and 3?	EB 39	Ann 10	Only step 2 has been applied.	OK	OK
k. In step 2, have all the sub-steps as below been	EB	Ann			

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
followed?	39	10			
i. Sub-step 2a: Determine appropriate analysis method;	EB 39	Ann 10	Yes.	OK	OK
ii. Sub-step 2b: Determine whether to apply simple cost analysis (Option I), investment comparison analysis (Option II) or benchmark analysis (Option III)	EB 39	Ann 10	Yes.	OK	OK
iii. Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III);	EB 39	Ann 10	Yes.	OK	OK
iv. Sub-step 2d: Sensitivity analysis (only applicable to Options II and III).	EB 39	Ann 10	Yes.	OK	OK
l. In sub-step 2a has the determination of appropraite method of analysis done as per the guidance as below?	EB 39	Ann 10			
i. Simple cost analysis if the CDM project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than CDM related income (Option I).	EB 39	Ann 10	Yes. The Project generates financial and economic benefits through the sales of electricity other than CDM related income therefore the simple cost analysis (Option I) cannot be taken.	OK	OK
ii. Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III). Specify option used with justification.	EB 39	Ann 10	Yes. Benchmark analysis (Option III) is applied.	OK	OK
m. Has the below guideline followed for sub-step 2b Option I. Apply simple cost analysis? Document the costs associated with the CDM project activity and the alternatives identified in Step 1 and	EB 39	Ann 10	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
demonstrate that there is at least one alternative which is less costly than the project activity.					
n. Has the below guideline followed for sub-step 2b Option II. Apply investment comparison analysis? Identify the financial indicator, such as IRR, NPV, cost benefit ratio, or unit cost of service most suitable for the project type and decision-making context. Please specify	EB 39	Ann 10	N/A.	OK	OK
o. Has the below guideline followed for Sub-step 2b: Option III. Apply benchmark analysis?	EB 39	Ann 10	Yes. Project IRR of 8% (post tax) is determined as the benchmark.	OK	OK
i. Identify the financial/economic indicator, such as IRR, most suitable for the project type and decision context.	EB 39	Ann 10	Yes. The project IRR excluding income tax is identified.	OK	OK
ii. When applying Option II or Option III, the financial/economic analysis shall be based on parameters that are standard in the market, considering the specific characteristics of the project type, but not linked to the subjective profitability expectation or risk profile of a particular project developer. Only in the particular case where the project activity can be implemented by the project participant, the specific financial/economic situation of the company undertaking the project activity can be considered.	EB 39	Ann 10	Yes. Project IRR of 8% (post tax) is widely used in the Chinese power sector. The financial analysis was based on parameters that are standard in the market, considering the specific characteristics of the project type, but not linked to the subjective profitability expectation or risk profile of the PP.	OK	OK
iii. Discount rates and benchmarks shall be derived from: (a) Government bond rates,	EB 39	Ann 10	Yes. The benchmark selection complies with option d: Government/official approved	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
increased by a suitable risk premium to reflect private investment and/or the project type, as substantiated by an independent (financial) expert or documented by official publicly available financial data; (b) Estimates of the cost of financing and required return on capital (e.g. commercial lending rates and guarantees required for the country and the type of project activity concerned), based on bankers views and private equity investors/funds' required return on comparable projects; (c) A company internal benchmark (weighted average capital cost of the company), only in the particular case referred to above in 2. The project developers shall demonstrate that this benchmark has been consistently used in the past, i.e. that project activities under similar conditions developed by the same company used the same benchmark; (d) Government/official approved benchmark where such benchmarks are used for investment decisions; (e) Any other indicators, if the project participants can demonstrate that the above Options are not applicable and their indicator is appropriately justified. Please specify benchmark and justify.			benchmark where such benchmarks are used for investment decisions. The benchmark was derived from *Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects* issued by former State Power Corporation of China in 2002, which has been used widely in feasibility studies of new power plants, including wind power projects in China.		
p. Has the below guideline followed for Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III)?	EB 39	Ann 10			

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
i. Calculate the suitable financial indicator for the proposed CDM project activity and, in the case of Option II above, for the other alternatives. Include all relevant costs (including, for example, the investment cost, the operations and maintenance costs), and revenues (excluding CER revenues, but possibly including inter alia subsidies/fiscal incentives, ODA, etc, where applicable), and, as appropriate, non-market cost and benefits in the case of public investors if this is standard practice for the selection of public investments in the host country.	EB 39	Ann 10	Yes, the IRR of the Project was calculated.	OK	OK
ii. Present the investment analysis in a transparent manner and provide all the relevant assumptions, preferably in the CDM-PDD, or in separate annexes to the CDM-PDD.	EB 39	Ann 10	The IRR calculation sheet has been provided to BVC.	OK	OK
iii. Justify and/or cite assumptions.	EB 39	Ann 10	Yes.	OK	OK
iv. In calculating the financial/economic indicator, the project's risks can be included through the cash flow pattern, subject to project-specific expectations and assumptions.	EB 39	Ann 10	Yes, the project's risks are included.	OK	OK
v. Assumptions and input data for the investment analysis shall not differ across the project activity and its alternatives, unless differences can be well substantiated.	EB 39	Ann 10	Not applicable as Option III is used.	OK	OK
vi. Present in the CDM-PDD a clear comparison of	EB	Ann	Yes, option III is used, and the IRR of the Project	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
the financial indicator for the proposed CDM activity.Please specify details for above.	39	10	is less than the benchmark of 8%.		
q. Has the below guideline followed for Sub-step 2d: Sensitivity analysis (only applicable to Options II and III)? Include a sensitivity analysis that shows whether the conclusion regarding the financial/economic attractiveness is robust to reasonable variations in the critical assumptions.	EB 39	Ann 10	Yes. The PDD includes fluctuation of four indicators from -10% to +10%, and states that the IRR could not reach the benchmark due to the impossibility of fluctuation of those indicators.	OK	OK
r. Has the outcome of Step 2 clearly mentioned with justification?	EB 39	Ann 10	~~Pending on CL-8:~~ Yes, the Project is financial unattractive without CDM revenues.	~~Pending~~	OK
s. In step 3: Barrier analysis have all the sub-steps as below been followed?	EB 39	Ann 10			
i. Sub-step 3a: Identify barriers that would prevent the implementation of the proposed CDM project activity;	EB 39	Ann 10	N/A.	OK	OK
ii. Sub-step 3b: Show that the identified barriers would not prevent the implementation of at least one of the alternatives (except the proposed project activity).	EB 39	Ann 10	N/A.	OK	OK
t. Has the below guideline followed for Sub-step 3a: Identify barriers that would prevent the implementation of the proposed CDM project?	EB 39	Ann 10			
i. (a) Investment barriers: For alternatives undertaken and operated by private entities: Similar activities have only been implemented with grants or other non-commercial finance terms. No private capital is available from	EB 39	Ann 10	N/A.	OK	OK

CHECKLIST QUESTION		Ref.	§	COMMENTS	Draft Concl	Final Concl
	domestic or international capital markets due to real or perceived risks associated with investment in the country where the proposed CDM project activity is to be implemented, as demonstrated by the credit rating of the country or other country investments reports of reputed origin.					
ii.	(b) Technological barriers: Skilled and/or properly trained labour to operate and maintain the technology is not available in the relevant country/region, which leads to an unacceptably high risk of equipment disrepair and malfunctioning or other underperformance; Lack of infrastructure for implementation and logistics for maintenance of the technology, Risk of technological failure: the process/technology failure risk in the local circumstances is significantly greater than for other technologies that provide services or outputs comparable to those of the proposed CDM project activity, as demonstrated by relevant scientific literature or technology manufacturer information, The particular technology used in the proposed project activity is not available in the relevant region.	EB 39	Ann 10	N/A.	OK	OK
iii.	(c) Barriers due to prevailing practice: The project activity is the “first of its kind”.	EB 39	Ann 10	N/A.	OK	OK
iv.	(d) Other barriers, preferably specified in the	EB	Ann	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
underlying methodology as examples.	39	10			
u. Has the outcome from Step 3a clearly mentioned in PDD?	EB 39	Ann 10	N/A.	OK	OK
v. Has the below guideline followed for Sub-step 3 b: Show that the identified barriers would not prevent the implementation of at least one of the alternatives (except the proposed project activity)?	EB 39	Ann 10			
i. If the identified barriers also affect other alternatives, explain how they are affected less strongly than they affect the proposed CDM project activity. In other words, demonstrate that the identified barriers do not prevent the implementation of at least one of the alternatives. Any alternative that would be prevented by the barriers identified in Sub-step 3a is not a viable alternative, and shall be eliminated from consideration.	EB 39	Ann 10	N/A.	OK	OK
ii. Provide transparent and documented evidence, and offer conservative interpretations of this documented evidence, as to how it demonstrates the existence and significance of the identified barriers and whether alternatives are prevented by these barriers.	EB 39	Ann 10	N/A.	OK	OK
iii. The type of evidence to be provided should include at least one of the following: (a) Relevant legislation, regulatory information or industry norms; (b) Relevant (sectoral) studies	EB 39	Ann 10	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
or surveys (e.g. market surveys, technology studies, etc) undertaken by universities, research institutions, industry associations, companies, bilateral/multilateral institutions, etc; (c) Relevant statistical data from national or international statistics; (d) Documentation of relevant market data (e.g. market prices, tariffs, rules); (e) Written documentation of independent expert judgments from industry, educational institutions (e.g. universities, technical schools, training centres), industry associations and others. Please specify.					
w. Has the outcome from Step 3 clearly mentioned in PDD?	EB 39	Ann 10	N/A.	OK	OK
x. In step 4: Common practise analysis have all the sub-steps as below followed?	EB 39	Ann 10			
i. Sub-step 4a: Analyze other activities similar to the proposed project activity;	EB 39	Ann 10	Yes.	OK	OK
ii. Sub-step 4b: Discuss any similar Options that are occurring.	EB 39	Ann 10	Yes.	OK	OK
y. Has the below guideline followed for Sub-step 4a: Analyze other activities similar to the proposed project activity? Provide an analysis of any other activities that are operational and that are similar to the proposed project activity. Other CDM project activities are not to be included in this analysis. Provide documented evidence and, where relevant, quantitative information. On the	EB 39	Ann 10	~~*CL-9:Please make an explanation that the geographical range of Heilongjiang Province and data source chosen in common practice analysis is appropriate in section B.5 of PDD.*~~ As per revised PDD, wind power projects located in Heilongjiang Province implemented after 2002 was identified as similar project and the similar	~~CL-9:~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
basis of that analysis, describe whether and to which extent similar activities have already diffused in the relevant region.			projects identified are complete.		
z. Has the below guideline followed for Sub-step 4b: Discuss any similar Options that are occurring? If similar activities are identified, then it is necessary to demonstrate why the existence of these activities does not contradict the claim that the proposed project activity is financially/economically unattractive or subject to barriers. This can be done by comparing the proposed project activity to the other similar activities, and pointing out and explaining essential distinctions between them that explain why the similar activities enjoyed certain benefits that rendered it financially/economically attractive (e.g., subsidies or other financial flows) and which the proposed project activity cannot use or did not face the barriers to which the proposed project activity is subject. In case similar projects are not accessible, the PDD should include justification about non-accessibility of data/information.	EB 39	Ann 10	~~Pending on CL-9:.~~ Yes. According to the Statistics of wind power installed capacity in China" (2007, 2008 and 2009), two wind power projects are identified as per the above criteria. As the two identified similar projects were demonstration projects and funded by international low interest loan or national soft loan. Thus they are essentially different from the Project.	~~Pending~~	OK
aa. Has the outcome from Step 4 clearly mentioned in PDD?	EB 39	Ann 10	~~Pending on CL-9:.~~ Yes, the project is not common practice.	~~Pending~~	OK
bb. Has it been proved that the porject is additional?	EB 39	Ann 10	Yes.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
a. Prior consideration of the clean development mechanism	VVM	98-104			
a. Is the project ativity start date prior to the date of publication of the PDD for stakeholder comments?	VVM	98	No. The project start date of 12/05/2009 is after the date of publication of the PDD for stakeholder comments of 12/09/2009.	OK	OK
b. If yes, were the CDM benefits considered necessary in the decision to undertake the project as a proposed CDM project activity?	VVM	98	Bureau Veritas Certification verified the PP's notification to China's DNA and the UNFCCC secretariat in writing of their intention to seek CDM status on 02/07/2009 and 22/07/2009 respectively. They were both made within six months of the start date of the Project.	OK	OK
c. Is the start date of the project activity, reported in the PDD, in accordance with the "Glossary of CDM terms", which states that "The starting date of a CDM project activity is the earliest date at which either the implementation or construction or real action of a project activity begins."?	VVM	99	Yes. 12/05/2009, the signed date of wind turbine purchase contract, was identified as the start date of the Project, which is the earliest date at which either the implementation or construction or real action of a project activity begins, hence it is in accordance with the latest CDM glossary.	OK	OK
d. Does the project activity require construction, retrofit or other modifications?	VVM	99	Yes, the Project requires construction.	OK	OK
e. If yes, is it ensured that the date of commissioning cannot be considered as the project activity start date?	VVM	99	No, the start date of the Project is the signed date of Wind Turbine Purchase agreement, which is not the date of commissioning.	OK	OK
f. Is it a new project activity (a project activity with a start date on or after 02 August 2008) or an existing project activity (a project activity with a start date before 02 August 2008)?	VVM EB 62	100 Ann 13	The start date of the Project is 12/05/2009, which is after 02/08/2008, so it is a new project activity	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
g. For a new project, for which PDD has not been published for global stakeholder consultation or a new methodology proposed to the CDM Executive Board before the project activity start date, had PPs informed the host Party DNA and the UNFCCC secretariat in writing of the commencement of the project activity and of their intention to seek CDM status? (Provide reference to such confirmation from host Party DNA and UNFCCC secretariat).	VVM EB 62	101 Ann 13	Yes, Bureau Veritas Certification verified the PP's notification to China's DNA and the UNFCCC secretariat in writing of their intention to seek CDM status on 02/07/2009 and 22/07/2009 respectively. They were both made within six months of the start date of the Project.	OK	OK
h. For an existing project activity, for which the start date is prior to the date of publication of the PDD for global stakeholder consultation, are the following evidences provided:	VVM EB 62	102 Ann 13			
i. evidence that must indicate that awareness of the CDM prior to the project activity start date, and that the benefits of the CDM were a decisive factor in the decision to proceed with the project, including, inter alia:	VVM EB 62	102 Ann 13	N/A.	OK	OK
a) minutes and/or notes related to the consideration of the decision by the Board of Directors, or equivalent, of the project participant, to undertake the project as a proposed CDM project activity?	VVM EB 62	102 Ann 13	N/A.	OK	OK
ii. reliable evidence from project participants that must indicate that continuing and real actions were taken to secure CDM status for the project in parallel with its	VVM EB 62	102 Ann 13			

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
implementation, including, inter alia:					
a) contract with consultants for CDM/PDD/methodology services?	VVM EB 62	102 Ann 13	N/A.	OK	OK
b) draft versions of PDDs	EB 62	Ann 13	N/A.	OK	OK
c) underlying documents such as letters of authorization	EB 62	Ann 13	N/A.	OK	OK
d) letters of intent	EB 62	Ann 13	N/A.	OK	OK
e) emission reduction purchase agreement (ERPA) term sheets	EB 62	Ann 13	N/A.	OK	OK
f) Emission Reduction Purchase Agreements or other documentation related to the sale of the potential CERs (including correspondence with multilateral financial institutions or carbon funds)?	VVM EB 62	102 Ann 13	N/A.	OK	OK
g) evidence of agreements or negotiations with a DOE for validation services?	VVM EB 62	102 Ann 13	N/A.	OK	OK
h) submission of a new methodology to the CDM Executive Board or requests for clarification or revision of existing methodologies to the CDM Executive Board?	VVM EB 62	102 Ann 13	N/A.	OK	OK
i) publication in newspaper?	VVM EB	102 Ann	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
	62	13			
j) interviews with DNA?	VVM EB 62	102 Ann 13	N/A.	OK	OK
k) earlier correspondence on the project with the DNA or the UNFCCC secretariat?	VVM EB 62	102 Ann 13	N/A.	OK	OK
l) If letters, e-mail exchanges and other documented communications help to substantiate the evidence, has the DOE assessed and confirmed the authenticity of such communications, inter alia through cross-checking (e.g. interviews).	EB 62	Ann 13	N/A.	OK	OK
m) Has the chronology of events including time lines been appropriately captured and explained/detailed in the PDD?	VVM EB 62	102 Ann 13	N/A.	OK	OK
b. Identification of alternatives	VVM	105- 107			
a. Does the approved methodology that is selected by the proposed CDM project activity prescribe the baseline scenario and hence no further analysis is required?	VVM	105	Yes. It has prescribed the baseline scenario i.e. Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
b. If no, does the PDD identify credible alternatives to the project activity in order to determine the most realistic baseline scenario?	VVM	105	Not applicable.	OK	OK
c. Does the list of alternatives given in the PDD esure that:	VVM	106			
i. the list of alternatives includes as one of the options that the project activity is undertaken without being registered as a proposed CDM project activity?	VVM	106	N/A. As the approved methodology ACM0002 selected by the proposed CDM project activity prescribes the baseline scenario and no further analysis is required.	OK	OK
ii. the list contains all plausible alternatives that the DOE, on the basis of its local and sectoral knowledge, considers to be viable means of supplying the outputs or services that are to be supplied by the proposed CDM project activity?	VVM	106	N/A.	OK	OK
iii. the alternatives comply with all applicable and enforced legislation?	VVM	106	N/A.	OK	OK
c. Investment analysis	VVM	108-114			
a. Has investment analysis been used to demonstrate the additionality of the proposed CDM project activity?	VVM	108	Yes.	OK	OK
b. If yes, does the PDD provide evidence that the proposed CDM project activity would not be:	VVM	108			
i. the most economically or financially	VVM	108	N/A. The benchmark analysis is applied.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
attractive alternative?					
ii. economically or financially feasible, without the revenue from the sale of certified emission reductions (CERs)?	VVM	108	Yes.	OK	OK
c. Was this shown by one of the following approaches?	VVM	109			
i. The proposed CDM project activity would produce no financial or economic benefits other than CDM-related income. Document the costs associated with the proposed CDM project activity and the alternatives identified and demonstrate that there is at least one alternative which is less costly than the proposed CDM project activity.	VVM	109	N/A.	OK	OK
ii. The proposed CDM project activity is less economically or financially attractive than at least one other credible and realistic alternative.	VVM	109	N/A.	OK	OK
iii. The financial returns of the proposed CDM project activity would be insufficient to justify the required investment.	VVM	109	Yes.	OK	OK
d. Is the period of assessment limited to the proposed crediting period of the CDM project activity?	EB 62	Ann 05	No. 21 years of assessment (20-year operation period and 1-year construction period) per the approved FSR.	OK	OK
e. Does the project IRR or equity IRR calculations reflect the period of expected operation of the underlying project activity (technical lifetime), or -	EB 62	Ann 05	The assessment of 21 years (1-year construction and 20-year operation) in project IRR calculation reflects the period of expected operation of the	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
if a shorter period is chosen - include the fair value of the project activity assets at the end of the assessment period?			Project.		
f. Does the IRR calculation include the cost of major maintenance and/or rehabilitation if these are expected to be incurred during the period of assessment?	EB 62	Ann 05	Yes, the major maintenance cost is included in IRR calculation.	OK	OK
g. Do the project participants justify the appropriateness of the period of assessment in the context of the underlying project activity, without reference to the proposed CDM crediting period?	EB 62	Ann 05	Yes. The period of assessment in the PDD is consistent with FSR.	OK	OK
h. Does the cash flow in the final year include a fair value of the project activity assets at the end of the assessment period?	EB 62	Ann 05	Yes. The revised IRR calculation sheet includes a fair value.	OK	OK
i. Has the fair value been calculated in accordance with local accounting regulations where available, or international best practice?	EB 62	Ann 05	Yes. The revised IRR calculation sheet includes a fair value. The fair value of 5% was selected reasonably following relevant regulation in China, i.e. The Document of the State Administration of Taxation, Decree No. 70 of Guoshuifa [2003] on 30 June 2003 and Statute of Income Tax Law of China issued on 06/12/2007.	OK	OK
j. Does the fair value calculations include both the book value of the asset and the reasonable expectation of the potential profit or loss on the	EB 62	Ann 05	Yes. The revised IRR calculation sheet includes a fair value.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
realization of the assets?			The fair value calculation can reasonably expect the potential profit recoverable.		
k. Was depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, added back to net profits for the purpose of calculating the financial indicator (e.g. IRR, NPV)?	EB 62	Ann 05	Yes, the depreciation, which has been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the IRR.	OK	OK
l. Has taxation been included as an expense in the IRR/NPV calculation in cases where the benchmark or other financial indicator is intended for post-tax comparisons?	EB 62	Ann 05	Yes. The income tax of 25% has been included in the IRR calculation.	OK	OK
m. Are the input values used in all investment analysis valid and applicable at the time of the investment decision taken by the project participant?	EB 62	Ann 05	Yes. The input values from the FSR were valid and applicable at the time of the investment decision.	OK	OK
n. Is the timing of the investment decision consistent and appropriate with the input values?	EB 62	Ann 05	Yes. The FSR was finalized in Sep. 2008 and approved on 24/03/2009. Later, based on the conclusion in the FSR, PP decided to invest the Project on 02/04/2009 with CDM consideration.	OK	OK
o. Are all the listed input values been consistently applied in all calculations?	EB 62	Ann 05	Yes.	OK	OK
p. Does the investment analysis reflect the economic decision making context at point of the decision to recomence the project in the case of project activities for which implementation ceases after the commencement and where	EB 62	Ann 05	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
implementation is recommenced due to consideration of the CDM?					
q. Have project participants supplied the spreadsheet versions of all investment analysis?	EB 62	Ann 05	Yes.	OK	OK
r. Are all formulas used in this analysis readable and all relevant cells be viewable and unprotected?	EB 62	Ann 05	Yes.	OK	OK
s. In cases where the project participant does not wish to make such a spreadsheet available to the public has the PP provided an exact read-only or PDF copy for general publication?	EB 62	Ann 05	N/A.	OK	OK
t. In case the PP wishes to black-out certain elements of the publicly available version, is it justifiable?	EB 62	Ann 05	N/A.	OK	OK
u. Was the cost of financing expenditures (i.e. loan repayments and interest) included in the calculation of project IRR?	EB 62	Ann 05	Yes.	OK	OK
v. In the calculation of equity IRR, has only the portion of investment costs which is financed by equity been considered as the net cash outflow?	EB 62	Ann 05	N/A.	OK	OK
w. Has the portion of the investment costs which is financed by debt been considered a cash outflow in the calcualtion of equity IRR? (this is not allowed)	EB 62	Ann 05	N/A.	OK	OK
x. Was a pre-tax benchmark be applied?	EB 62	Ann 05	No. A post-tax benchmark is applied.	OK	OK
y. In cases where a post-tax benchmark is applied, is actual interest payable taken into account in	EB 62	Ann 05	A post-tax benchmark is applied in the investment analysis.	~~CAR 6:~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
the calculation of income tax?			~~*CAR-6:As a post tax benchmark is applied, the actual interest payable should be taken into account in the calculation of income tax in the IRR calculation.*~~ The actual interest payable has been taken into account in the calculation of income tax in the revised IRR calculation sheet, and the interest rate and debt-equity ratio has been checked as consistent with the approved FSR.		
z. In cases where a benchmark approach is used is the applied benchmark appropriate to the type of IRR calculated?	EB 62	Ann 05	Yes.	OK	OK
aa. Has local commercial lending rates or weighted average costs of capital (WACC) selected as appropriate benchmarks for a project IRR?	EB 62	Ann 05	No.	OK	OK
bb. Has required/expected returns on equity selected as appropriate benchmark for an equity IRR?	EB 62	Ann 05	No.	OK	OK
cc. In case benchmarks supplied by relevant national authorities selected is it applicable to the project activity and the type of IRR calculation presented?	EB 62	Ann 05	Yes. The benchmark was source from *Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects* issued by former State Power Corporation of China in 2002, which has been used widely in feasibility studies of new power plants, including wind power projects in China.	OK	OK
dd. In the cases of projects which could be developed by an entity other than the project participant is the benchmark applied based on	EB 62	Ann 05	No.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
parameters that are standard in the market?					
ee. Whether a company-specific benchmark or a benchmark based on parameters that are standard in the market is suitable in the context of the underlying project activity?	EB 62	Ann 05	The standard in the market is applied. The benchmark was source from *Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects* issued by former State Power Corporation of China in 2002, which has been used widely in feasibility studies of new power plants, including wind power projects in China.	OK	OK
ff. Have internal company benchmarks/expected returns (including those used as the expected return on equity in the calculation of a weighted average cost of capital - WACC) been applied in cases where there is only one possible project developer?	EB 62	Ann 05	No.	OK	OK
gg. In such cases, have these values been used for similar projects with similar risks, developed by the same company or, if the company is brand new, would have been used for similar projects in the same sector in the country/region?	EB 62	Ann 05	N/A.	OK	OK
hh. Has a minimum clear evidence of the resolution by the company's Board and/or shareholders been provided to the effect as above?	EB 62	Ann 05	N/A.	OK	OK
ii. Has a thorough assessment of the financial statements of the project developer - including the proposed WACC - to assess the past financial behavior of the entity during at least the last 3 years in relation to similar projects been conduted?	EB 62	Ann 05	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
jj. If a company internal benchmark is used, is it derived from the Capital Asset Pricing Model (CAPM)?	EB 62	Ann 05	No.	OK	OK
kk. If yes, are the resulting benchmarks consistently used by the company in the past?	EB 62	Ann 05	N/A.	OK	OK
ll. If the benchmark is based on parameters that are standard in the market, is the cost of equity determined either by: (a) selecting the values provided in Appendix A; or by (b) calculting the cost of equity using best financial practices, based on data sources which can be clearlyvalidated by the DOE, while properly justifying all underlying factors.	EB 62	Ann 05	No.	OK	OK
mm. If a company internal benchmark is used, are the values in the table in Appendix A used, as a simple default option?	EB 62	Ann 05	N/A.	OK	OK
nn. If a company's internal benchmark is used for the expected return on equity, is the cost of debt based on the weighted average cost of debt financing of the legal entity owning the CDM project activity?	EB 62	Ann 05	N/A.	OK	OK
oo. For loans, is the weighted average cost of outstanding long-term debt used?	EB 62	Ann 05	N/A.	OK	OK
pp. For bonds, is the weighted average yield of the bonds during the last three months prior to the submission of the CDM-PDD for validation or prior to the investment decision, whichever is earlier, used? The use of bonds to determine the	EB 62	Ann 05	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
cost of debt is only appropriate for corporate bonds issued in the host country of the CDM project.					
qq. In cases where the debt finance structure of the project is not yet available (e.g. a letter of intent for debt funding is not available), the cost of debt can be assumed as the commercial lending rate in the country or the yield of a 10 year bond issued by the government of the host country or, if this is not available, the bond with the maturity which is closest to 10 years. The following should be documented in the CDM-PDD:	EB 62	Ann 05	N/A.	OK	OK
rr. (a) for bonds: the key parameters of the bond including the time of maturity, yield, registration issuance in the financial system and set-up in the market;	EB 62	Ann 05	N/A.	OK	OK
ss. (b) for loans from a financial institution: the contract of lending between the financial institution and the legal entity owning the assets of the project activity, or, in absence of the contract, a letter from the bank stating its intention to award the loan and the key terms for the loan;	EB 62	Ann 05	N/A.	OK	OK
tt. for debt financing from a parent company: the transfer of capital to the legal entity, documented with the contract of lending between the parent company and the legal entity owning the assets of the project activity and/or the parameters of	EB 62	Ann 05	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
the corporate bonds as mentioned above.					
uu. This latter option is only valid for corporate bonds issued in the host country of the CDM project activity.	EB 62	Ann 05	N/A.	OK	OK
vv. If the benchmark is based on parameters that are standard in the market, is the cost of debt e calculated as the cost of financing in the capital markets (e.g. commercial lending rates and guarantees required for the country and the type of project activity concerned), based on documented evidence from financial institutions with regard to the cost of debt financing of comparable projects?	EB 62	Ann 05	N/A.	OK	OK
ww. In cases where this data is not available, is the commercial lending rate in the host country used to calculate the cost of debt.	EB 62	Ann 05	N/A.	OK	OK
xx. If a company's internal benchmark is used for the expected return on equity, is the percentage of debt financing and equity financing reflect the long-term debt/equity finance structure of the legal entity owning the assets of the project activity?	EB 62	Ann 05	N/A.	OK	OK
yy. If: (a) the legal entity owning the assets of the project activity has balance sheets audited by a third party within two years prior to the submission of the CDM-PDD for validation; and (b) the accounting books of the legal entity reflect at least the total value of all the assets needed	EB 62	Ann 05	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
for the project activity. Is the percentage determined based on the latest balance sheet provided under local fiscal/accounting standards and rules?					
zz. If the debt/equity finance structure is not yet available, 50% debt and 50% equity financing may be assumed as a default.	EB 62	Ann 05	N/A.	OK	OK
aaa. Is the benchmark based on parameters that are standard in the market?	EB 62	Ann 05	N/A.	OK	OK
bbb. If yes, is the typical debt/equity finance structure observed in the sector of the country used? ccc.	EB 62	Ann 05	N/A.	OK	OK
ddd. If such information is not readily available, 50% debt and 50% equity financing may be assumed as a default.	EB 62	Ann 05	N/A.	OK	OK
eee. Has an investment comparison analysis and not a benchmark analysis used when the proposed baseline scenario leaves the project participant no other choice than to make an investment to supply the same (or substitute) products or services?	EB 62	Ann 05	N/A.	OK	OK
fff. Have variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues been subjected to reasonable variation (positive and negative) and the results of this variation been presented in the PDD and be reproducible in the associated spreadsheets?	EB 62	Ann 05	The sensitivity analysis for four indicators, i.e. total static investment, annual O&M cost, annual grid connected output and tariff, with range from -10% to +10% has been provided, and it states that the IRR could not reach the benchmark due to the impossibility of fluctuation of those indicators.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
ggg. Have a corrective action been raised for a variable to be included in the sensitivity analysis which constitute less than 20% and have a material impact on the analysis ?	EB 62	Ann 05	No.	OK	OK
hhh. Is the range of variations selected is reasonable in the project context?	EB 62	Ann 05	Yes. The range of -10%-+10% has been analyzed.	OK	OK
iii. Does the variations in the sensitivity analysis at least cover a range of +10% and -10%, unless this is not deemed appropriate in the context of the specific project circumstances?	EB 62	Ann 05	Yes.	OK	OK
jjj. In cases where a scenario will result in the project activity passing the benchmark or becoming the most financially attractive alternative, is an assessment done of the probability of the occurrence of this scenario in comparison to the likelihood of the assumptions in the presented investment analysis, taking into consideration correlations between the variables as well as the specific socio-economic and policy context of the project activity?	EB 62	Ann 05	The PDD includes information about fluctuation of those indicators, and states that the IRR could not reach the benchmark due to the impossibility of fluctuation of those indicators.	OK	OK
kkk. Was the plant load factor defined ex-ante in the CDM-PDD according to one of the following options:	EB 48	Ann 11			
i. The plant load factor provided to banks and/or equity financiers while applying the project activity for project financing, or to the government while applying the project activity for implementation approval?	EB 48	Ann 11	Yes. The PLF in the FSR is the one was applied for local government approval.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
ii. The plant load factor determined by a third party contracted by the project participants (e.g. an engineering company)?	EB 48	Ann 11	Yes. The PLF is determined by an authorized third part contracted by the PP.	OK	OK
lll. Was a thorough assessment of all parameters and assumptions used in calculating the relevant financial indicator, and determine the accuracy and suitability of these parameters using the available evidence and expertise in relevant accounting practices conducted?	VVM	111	The revised PDD includes information about fluctuation of those indicators, and states that the IRR could not reach the benchmark due to the impossibility of fluctuation of those indicators.	OK	OK
mmm. Were the parameters cross-checked against third-party or publicly available sources, such as invoices or price indices?	VVM	111	Yes. The parameters have been checked with the actual regulations. The operation period of 20 years were selected reasonably following the requirements of “Interim Rules on Economic Assessment of Electric Power Engineering Retrofit Projects”. The total static investment in the approved FSR is 482.69 million RMB is in the range of the registered CDM wind power projects in Heilongjiang Province. And the actual static total investment is 1.3% higher than the value in the FSR. The board decision was made on 02/04/2009, at which tariff notification Fa Gai Jia Ge [2008] 1876 and Fa Gai Jia Ge [2008] 1678 were available after the implementation of Renewable Energy Law. Therefore, BVC can confirm that 0.61RMB/kWh (incl. VAT) in the first 30,000	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			operation hours and 0.41RMB/kWh (incl. VAT) after 30,000 operation hours employed in investment analysis is available and appropriate at the time of board decision. The supplied electricity of the Project was based on the 30-year long term wind resource history data of the local area, and WAsP software as stated in approved FSR.Therefore the supplied electricity is found appropriate. BVC confirms that the annual O&M cost is the sum of salary and welfare of employees, materials fee, insurance cost, maintenance fee and miscellaneous account, which was studied based on the “Code on Compiling Feasibility Study Report of Wind Farms” issued by NDRC and “Economic Evaluation Method and Parameters for Project Construction” (version 3) BVC has checked the loan contract signed between the PP and the bank, and found that the Project actually enjoys 10% discount on the interest rate and the dept rate is 25%. As a post tax benchmark is applied, the actual interest payable should be taken into account according to “Guidelines on the assessment of investment analysis” version 05. The residual value rate of 5% was selected reasonably following relevant regulation in China, i.e. The Document of the State Administration of		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Taxation, Decree No. 70 of Guoshuifa [2003] on 30 June 2003 and Statute of Income Tax Law of China issued on 06/12/2007 The VAT of 8.5% complies with the Notice of Value added Tax Policy Regarding Products Using Certain Synthesized Resources and Other Products issued by the Ministry of Finance and the State Administration of Taxation in Dec. 2001 at the time of FSR. The income tax of 25% complies with Enterprise Income Tax Law of China which is effective from 01/01/2008		
nnn. Were feasibility reports, public announcements and annual financial reports related to the proposed CDM project activity and the project participants reviewed?	VVM	111	Yes.	OK	OK
ooo. Was the correctnes of computations carried out and documented by the project participants assessed?	VVM	111	Yes.	OK	OK
ppp. Was the sensitivity analysis by the project participants to determine under what conditions variations in the result would occur, and the likelihood of these conditions assessed?	VVM	111	In the PDD, all parameters and assumption used in calculating the relevant financial indicator are accurate and suitable.	OK	OK
qqq. Is the type of benchmark applied suitable for the type of financial indicator presented?	VVM	112	Yes. The benchmark of 8% is widely used for wind power projects similar to the Project in China.	OK	OK
rrr. Do any risk premiums applied determining the benchmark reflect the risks associated with the	VVM	112	N/A. The benchmark of 8% is widely used for wind	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
project type or activity?			power projects similar to the Project in China.		
sss. To determine this, was it assessed whether it is reasonable to assume that no investment would be made at a rate of return lower than the benchmark by:	VVM	112			
i. assessing previous investment decisions by the project participants involved?	VVM	112	N/A.	OK	OK
ii. determining whether the same benchmark has been applied?	VVM	112	N/A.	OK	OK
iii. determining if there are verifiable circumstances that have led to a change in the benchmark?	VVM	112	N/A.	OK	OK
ttt. Did the project participants rely on values from Feasibility Study Reports (FSR) that are approved by national authorities for proposed CDM project activities?	VVM	113	Yes.	OK	OK
aaa. If yes:	VVM	113			
i. has the FSR been the basis of the decision to proceed with the investment in the project, i.e. that the period of time between the finalization of the FSR and the investment decision is sufficiently short for the DOE to confirm that it is unlikely in the context of the underlying project activity that the input values would have materially changed?	VVM	113	Yes, as interviewed, the PP's final decision to proceed with the investment in the Project has been made based on the FSR finalized in Sep. 2008 and decided to invest the project soon on 02/04/2009 with consideration of CDM revenues. The validation team was therefore confident that it is unlikely in the context of the underlying project activity that the input values would have materially changed.	OK	OK
ii. Are the values used in the PDD and associated annexes fully consistent with the	VVM	113	No.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
FSR?			Except the actual interest payable, all other input parameters are sourced from the approved FSR.		
iii. If not, was the appropriateness of the values validated?	VVM	113	Yes. As a post tax benchmark is applied, the actual interest payable should be taken into account in the calculation of income tax in the IRR calculation.	OK	OK
iv. On the basis of its specific local and sectoral expertise, is confirmation provided, by cross-checking or other appropriate manner, that the input values from the FSR are valid and applicable at the time of the investment decision?	VVM	113	Yes, the total investment was crosschecked with the signed contract and Final Accounting Report. The PLF was crosschecked with the approved FSR and the registered wind projects in Heilongjiang Province. The annual O&M cost was crosschecked with the registered wind projects in Heilongjiang Province. And the tariff was crosschecked with relevant tariff notifications.	OK	OK
d. Barrier analysis	VVM	115-118			
a. Has barrier analysis been used to demonstrated the additionality of the proposed CDM project activity?	VVM	115	No.	OK	OK
b. If yes, does the PDD demonstrate that the proposed CDM project activity faces barriers that:	VVM	115			
i. prevent the implementation of this type of proposed CMD project activity?	VVM	115	N/A.	OK	OK
ii. do not prevent the implementation of at least one of the alternatives?	VVM	115	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
c. Are there any issues that have a clear direct impact on the financial returns of the project activity, other than: risk related barriers, for example risk of technical failure, that could have negative effects on the financial performance; or barriers related to the unavailability of sources of finance for the project activity? {If yes, these issues cannot be considered barriers and shall be assessed by investment analysis. [Refer to (6.c) above]}	VVM	116	N/A.	OK	OK
d. Were the barriers determined as real by:	VVM	117			
i. assssing the available evidence and/or undertaking interviews with relevant individuals (including members of industry associations, government officials or local experts if necessary) to determine whether the barriers listed in the PDD exist?	VVM	117	N/A.	OK	OK
ii. ensuring that existence of barriers is substantiated by independent sources of data such as relevant national legislation, surveys of local conditions and national or international statistics?	VVM	117	N/A.	OK	OK
iii. Is existence of a barrier substantiated only by the opinions of the project participants? (If yes, this barrier cannot be considered as adequately substantiated)	VVM	117	N/A.	OK	OK
e. Were the barriers determined as preventing the implementation of the project activity but not the	VVM	117	N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
implementation of at least one of the possible alternatives by applying local and sectoral expertise to judge whether a barrier or set of barriers would prevent the implementation of the proposed CDM project activity and would not equally prevent implementation of *at least one of* the possible alternatives, in particular the identified baseline scenario?					
e. Common practice analysis	VVM	119-121			
a. Is this a proposed large-scale, or first-of-its kind small-scale project activity?	VVM	119	The Project is a large scale project.	OK	OK
b. If yes, was common practice analysis carried out as a credibility check of the other available evidence used by the project participants to demonstrate additionality?	VVM	119	~~Pending on CL-9:.~~ The revised common practice is considered complete.	~~Pending~~	OK
c. Was it assessed whether the geograpphical scope (e.g. defined region) of the common practice analysis is appropriate for the assessment of common practice related to the project activity's technology or industry type? (For certain technologis the relevatn region for assessment will be local and for others it may be transnational/global.	VVM	120	Heilongjiang Province is selected as the geographical scope which is large enough for the assessment of common practice.	OK	OK
d. Was a region other than the entire host country chosen?	VVM	120	Yes, Heilongjiang Province is chosen.	OK	OK
e. If yes, was the explanation why this region is more appropriate assessed?	VVM	120	Yes, Heilongjiang Province, which is the connected grid of the Project, is large enough for	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			analysis.		
f. Using official sources and local and industry expertise, was it determined to what extent similar and operational projects (e.g., using similar technology or practice), other than CDM project activities, have been undertaken in the defined region?	VVM	120	~~Pending on CL-9:.~~ China Windfarm Capacity Statistic 2007, 2008 and 2009 is applied.	~~Pending~~	OK
g. Are similar and operational projects, other than CDM project activities, already "widely observed and commonly carried out" in the defined region?	VVM	120	~~Pending on CL-9:.~~ Two similar project activities were identified and considered complete.	~~Pending~~	OK
h. If yes, was it assessed whether there are essential distinctions between the proposed CDM project activity and the other similar activities?	VVM	120	~~Pending on CL-9:.~~ As the two identified similar projects were demonstration projects and funded by international low interest loan or national soft loan.	~~Pending~~	OK
7. Monotoring plan	VVM	122-124			
a. Does the PDD include a monitoring plan?	VVM	122	Yes.	OK	OK
b. Is this monitoring plan based on the approved monitoring methodology applied to the proposed CDM project activity?	VVM	122	Yes.	OK	OK
c. Were the list of parameters required by the the selected methodology identified?	VVM	123	~~Pending on CL-4:.~~ Electricity supplied by the Project to the grid and Electricity imported by the Project from the grid will be monitored to get the net electricity	~~Pending~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			generated.		
d. Does the monitoring plan contains all necessary parameters?	VVM	123	~~Pending on CL-4:.~~ Electricity supplied by the Project to the grid and Electricity imported by the Project from the grid will be monitored to get the net electricity generated.	~~Pending~~	OK
e. Are the parameters clearly described?	VVM	123	~~Pending on CL-4:.~~ Electricity supplied by the Project to the grid and Electricity imported by the Project from the grid will be monitored to get the net electricity generated.	~~Pending~~	OK
f. Does the means of monitoring described in the plan comply with the requirements of the methodology?	VVM	123	~~Pending on CL-4:.~~ Electricity supplied by the Project to the grid and Electricity imported by the Project from the grid will be monitored to get the net electricity generated.	~~Pending~~	OK
g. Are the monitoring arrangements described in the monitoring plan feasibl within the project design?	VVM	123	Yes.	OK	OK
h. Are the following means of implementation of the monitoring plan sufficient to ensure that the emission reductions achieved by/resulting from the proposed CDM project activity can be reported ex post and verified:	VVM	123			
i. data management procedures?	VVM	123	~~Pending on CL-4:.~~	~~Pending~~	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			The revised monitoring plan is considered complete and sufficient.		
ii. quality assurance procedures?	VVM	123	Yes.	OK	OK
iii. quality control procedures?	VVM	123	Yes.	OK	OK
8. Sustainable development	VVM	125-127			
a. Does the CDM project activity assists Parties not included in Annex I to the Convention in achieving sustainable development?	VVM	125	Yes.	OK	OK
b. Does the letter of approval by the DNA of the host Party confirm the contribution of the proposed CDM project activity to the sustainable development of the host Party?	VVM	126	Yes.	OK	OK
9. Local stakeholder consultation	VVM	128-130			
a. Were local stakeholders (public, including individuals, groups or communities affected, of likely to be affected, by the proposed CDM project activity or actions leading to the implementation of such an activity) invited by the PPs to comment on the proposed CDM project activity prior to the publication of the PDD on the UNFCCC website?	VVM	128	Yes. The local stakeholders were invited by the PP in May 2009; PP conducted surveys to stakeholders including local residents and governments. 10 stakeholder representatives were invited in a symposium on 06/05/2009 and all the stakeholder representatives support and welcome the proposed project. Then total 40 questionnaires had been distributed in nearby	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			areas.		
b. Have comments by local stakeholders that can reasonably be considered relevant for the proposed CDM project activity been invited?	VVM	129	Yes.	OK	OK
c. Is the summary of the comments received as provided in the PDD complete?	VVM	129	~~Pending on CL-6:~~ The revised summary in the PDD is complete.	~~Pending~~	OK
d. Have the project participants taken due account of any comments received and described this process in the PDD?	VVM	129	As Section E.3.of PDD addressed, there is no need to be modified the project due to the comments received.	OK	OK
10. Environmental impacts	VVM	131-133			
a. Have the project participants submitted documentation on the analysis of the environmental impacts of the project activity?	VVM	131	Yes. EIA and its approval made by local EPA are presented.	OK	OK
b. Have the project participants undertaken an analysis of environmental impacts?	VVM	132	Yes.	OK	OK
c. Does the host Party require an environmental impact assessment?	VVM	132	Yes.	OK	OK
d. If yes, have the project participants undertaken an environmental impact assessment?	VVM	132	Yes. Approved on 25/07/2008 by local EPA	OK	OK

Table 2 Resolution of Corrective Action and Clarification Requests

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
CAR-1: LoA from China should be provided.	1	The LoA from China's DNA has been provided on 08/08/2011, which was issued by China NDRC on 30/01/2010.	The LoA from China has been checked, so CAR-1: is closed out.
CAR-2: LoA from the Annex I Country should be provided.	1	The LoA from DNA of the buyer has been provided on 17/10/2011, which was issued by UK's DNA on 04/10/2011.	The LoA from UK has been checked, so CAR-2: is closed out.
CAR-3: As per the latest CDM PDD guideline, the lifetime of the equipments based on manufacturer's specifications and load factor should be listed in section A.4.3 of PDD. In accordance with the FSR, operational period of the Project is 20 years. A correction in Section B.5 and C.1.2 should be provided.	3.g.iii	The lifetime of the equipments based on manufacturer's specifications and load factor has been added in the revised PDD. Section B.5 and C.1.2 have been revised to be consistent with the FSR and manufacturer's specifications.	The revised part has been verified as consistent with the provided evidence, so CAR-3: is closed out.
CAR-4: The latest "Tool to calculate the emission factor for an electricity system" version 02.2.1 should be used.	3.j.i	The PDD has been revised according to the EF tool version 02.2.1. Details have shown in PDD, part B.6.1.	The relevant part of the PDD has been verified as consistent with the EF tool version 02.2.1, so CAR-4: is closed out.
CAR-5: It is stated in section B.6.1. of the PDD that there	5.e.c	According to the official statistic and the calculation issued on 02/07/2009, the relevant sections have been revised	The part of PDD has been revised and consistent with the

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
were electricity imports from NWPG to NECPG. However, from the official statistic and the calculation issued on 02/07/2009 by Chinese DNA, there were no electricity imports from NWPG to NECPG. Please correct relevant description in the PDD.		and updated.	official statistic and calculation. So CAR-5: is closed out.
CAR-6: As a post tax benchmark is applied, the actual interest payable should be taken into account in the calculation of income tax in the IRR calculation.	6.y	In the approved FSR, the Project IRR didn't take payable interest into account, according to China's national guidance, i.e., the Methodology and Parameters of Economic Evaluation on Construction Projects. For the proposed project, the actual loan from the bank was only 25% of static total investments with the favourable rate of long-term loan about 5.346%. The reason was that the bank estimated the project and thought the proposed project had a big risk to repay the loan because of the low IRR. Therefore, according to GUIDELINES ON THE ASSESSMENT OF INVESTMENT ANALYSIS (version 05), IRR calculation has been revised by taking payable interest into account based on the data from Loan letter issued by Heilongjiang Branch bank of China Construction Bank. Details have been shown in IRR calculation sheet.	As a post tax benchmark is applied, the actual interest rate and equity-dept rate has been applied. It is in line with GUIDELINES ON THE ASSESSMENT OF INVESTMENT ANALYSIS, and 5.346% is 90% discount of 5.94% which is the same as prevailing commercial interest rates in China, so CAR-6: is closed out.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
		The IRR was changed from 6.26% to 6.81%.	
CL-1: The emissions sources and GHGs involved should be clarified in section A.4.3	3.g.iv	The Project will supply an average annual generation of 120,069MWh to NECPG and replace the same amount of electricity generated by fossil fuel fired power plants connected to NECPG to reduce greenhouse gas emissions of CO_2	The emissions sources and GHGs involved have been added, so CL-1: is closed out.
CL-2: ACM0002 version 12.1.0 was valid from 26/11/2010. The applied version of methodology should be updated.	3.j.i	The ACM0002 has been updated according to the latest version (12.1.0). the relevant sections have been revised in the PDD version 02.	ACM0002 version 12.1.0 has been applied in the revised PDD, so CL-2: is closed out.
CL-3: The baseline scenario has been identified directly in ACM0002. Please explain the baseline scenario identification in accordance with ACM0002 version 12.1.0 in section B.4. of the PDD.	3.m.i	According to the description on the application of ACM0002 below: *Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".* The proposed project is connected to Heilongjiang Province Grid, which is an integrated part of Northeast China Power Grid (NECPG). Therefore, the baseline scenario of the proposed project is provision of equivalent amount of annual power output by the NECPG where the proposed project is connected, which is the continued operation of the existing power plants and the addition of	The revised baseline scenario and it identification are in line with the applied methodology, so CL-3: is closed out.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
		new generation sources on the NECPG to meet the electricity demand.	
CL-4: The monitoring information about how the electricity exported and imported will be monitored should be provided in section B.7.1 of the PDD.	3.s.i	The relevant descriptions about how to monitor the electricity exported and imported have been added in section B7.1 of the PDD.	The revised monitoring parameters are considered complete. So CL-4: is closed out.
CL-5: Please clarify calibrating frequency of the monitoring meter and the emergency measures that will be taken when the meters cannot work normally in section B.7.2 of the PDD.	3.t.i	The metering equipments at the substation are calibrated and tested yearly by a qualified third party appointed by the Northeast China Power Grid for accuracy according to the requirement from Technical administrative code of electric energy metering. The quality assurance and quality control procedures for the emergency about recording, maintaining and archiving data shall be improved as part of this CDM project activity. This is an on-going process that will be ensured through the CDM in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002. Hegang Longyuan Wind Power Co., Ltd specially issued the regulations of monitoring and management for CDM project (hereafter refer as CDM Manual) to monitor and verify the emission reductions from the proposed project. Moreover, the management procedures and regulations like operation, meter recordings, maintenance and emergency management are established in the CDM	The revised MP is considered complete, so CL-5: is closed out.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
		Manual, which guides the technicians to operate and guarantee the quality assurance and quality control procedures for recording, maintaining and archiving data in detail. This is an on-going process that will be ensured through the CDM in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002. Several processes required to meet the requirements for emissions reduction monitoring include: The project owner reads the meters system monthly and supplies the recordings to the local Electric Power Company. The local Electric Power Company checks the recordings supplied by the project owner. If either party finds any reading of the Meters more than the allowable error or the monitoring equipment functions improperly, it should inform the other party immediately. The project owner and the local Power Grid Company should retain a qualified measure institute together to check the meters or equipment, solve the problems and get everything into normal condition; In the mean time, both sides shall jointly prepare an acceptable estimation method according to Regulations and Rules of Power Supply issued by State Electricity Regulatory Commission of China to calculate the electricity exported to the Grid, and provide evidence to DOE for the verification to show the estimation is reasonable and conservative.	

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
		For the handling of disputes between the proposed project owner and the Grid, measures will be adopted according to relevant articles of Interim Measures for Settlement of Electricity Charges between the Power Generating Enterprises and the Grid Enterprises issued by State Electricity Regulatory Commission of China. The descriptions about calibrating frequency of the monitoring meter and the measures to solve the emergency situation of monitoring system have been clarified in detail in section B7.2 of the revised PDD.	
CL-6: The summary of the comments received is not complete.	3.aa.ii	To check the original references, there were 9 questions on each questionnaire. In the last version PDD, a minor part of the survey results of the questionnaire had been overlooked. In the PDD version 2.0, the total survey results of 9 questions on each questionnaire have been provided respectively in section E.2 of PDD. Summary views on the questionnaires: ✧ 40 respondents (100%) know the proposed project ✧ 40 respondents (100%) think that the proposed project brings positive impacts to local economy development. ✧ 40 respondents (100%)think the proposed project	The revised summary of the comments received is consistent with the questionnaires and complete, so CL-6: is closed out.

Draft report clarifications and corrective action requests by validation team	**Ref. to checklist question in table 1**	**Summary of project owner response**	**Validation team conclusion**
		bring positive impacts to incensement of employment opportunities ✧ 40 respondents (100%) think that the proposed project brings mild impacts to of local environment during the construction period? ✧ 40 respondents (100%) think that the proposed project brings mild impacts to of local environment during the operation period. ✧ 40 respondents (100%) think that bring several positive impacts to local residents' living, such as lessening of power shortage; increasing of income; improvement standard of living; improvement of local environment; increasing of employment opportunities. ✧ 12 respondents (30%) think that bring several negative impacts to local residents' living, such as impacts of water quality; ecological environment; construction noise; quality of the air. ✧ 40 respondents (100%) support the construction of the proposed project. 40 respondents (100%) don't have other advices for the proposed project except the lists on the questionnaire.	
CL-7: Evidences should be provided to prove there are no adequate biomass sources,	6.f	More evidences have been provided to explain that biomass sources, solar sources, wave and tidal sources or geothermal sources could not be applied in the project	The evidences and discussion has been verified, and Alternative III is correctly

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
solar sources, wave and tidal sources or geothermal sources for constructing a power plant with the same grid power as the Project.		site. In B5 of the PDD, the relevant descriptions on the alternatives have been updated.	exculed. So CL-7: is closed out.
CL-8: Please clearly clarify the outcome of Step 2 in section B.5. of the PDD according to the latest version of '*Tool for The Demonstration and Assessment of Additionality*'.	6.r	According to "*Tool for The Demonstration and Assessment of Additionality*", the clear outcome of Step 2 in section B5 of the PDD has been revised.	The outcome of Step 2 in section B.5 is clear, so CL-8: is closed out.
CL-9: Please make an explanation that the geographical range of Heilongjiang Province and data source chosen in common practice analysis is appropriate in section B.5 of PDD.	6.y	According to the definitions of other activities similar to the proposed project activity in "Tool for the Demonstration and Assessment of Additionality", the similar projects are considered similar if they are in the same country/region and/or rely on a broadly similar technology, are of a similar scale, and take place in a comparable environment with respect to regulatory framework, investment climate, access to technology, access to financing, etc. The activities similar to the proposed project activity were selected according to the following three rules: 1. The activities should locate in the same province. Firstly, the wind power project was implemented under the administration of provincial level government. Secondly,	The revised criteria and data source are considered suitable, so CL-9: is closed out.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
		the activities in the same province have the similar wind resource, grid structure, geological and economic developing conditions. The former two factors affected the estimated average annual output and the later affected the total investment respectively. Consequently, the similar activities should be limited within Heilongjiang Province which the proposed projects located. 2. The similar activities should be implemented after 2002. Before 2002, Chinese Power System had not been reformed and power grid and power generation had not been separated. There was no bidding process was required and power companies and grid companies share the same interests. The Power System reform happened in 2002 in China. As a result, new power project, including wind power, was expected to compete under more commercial conditions. Moreover, the Chinese Government launched the Wind Bidding Mechanics. Accordingly, the wind power projects were developed in the different regulatory condition after 2002. 3. Following the clarification made in the EB38 meeting report (paragraph 60), project activities which have been registered or validated as CDM projects on the UNFCCC CDM website are not identified as similar activities to the proposed project. The revised section has been added in section B.5 of PDD according to the latest rules issued by EB.	

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1	Summary of project owner response	Validation team conclusion
		The wind project statistics 2007, 2008 and 2009 have been used.	